

## RESEARCH ARTICLE OPEN ACCESS

# Evaluating the Influence of Board Characteristics on Environmental Decoupling: Evidence From Europe

Sabrina Pisano<sup>1</sup>  | Luigi Lepore<sup>1</sup>  | Raffaella Nastari<sup>1</sup> | Bakr Al-Gamrh<sup>2</sup> 

<sup>1</sup>Parthenope University of Naples, Naples, Italy | <sup>2</sup>Rennes School of Business, Rennes, France

**Correspondence:** Bakr Al-Gamrh ([bakr.ali@rennes-sb.com](mailto:bakr.ali@rennes-sb.com))

**Received:** 31 July 2024 | **Revised:** 28 April 2025 | **Accepted:** 5 May 2025

**Keywords:** agency theory | board gender diversity | corporate social responsibility | CSR committee | disclosure–performance gap | independent board member

## ABSTRACT

This study investigates the relationship between corporate governance characteristics and environmental decoupling, that is, the misalignment between environmental disclosure and environmental performance. We analyze a sample of 728 European companies (3061 firm-year observations) belonging to 18 industries and 20 different countries from 2017 to 2023. The results show that companies tend not to disclose all the environmental actions implemented, indicating underreporting behavior. The results also reveal that board independence, board gender diversity, and the presence of a CSR committee mostly foster a reduction in environmental decoupling. Furthermore, these corporate governance characteristics are also found to be effective mechanisms in enhancing companies' environmental performance. However, only the presence of a CSR committee has a strong positive effect on the quantity of environmental information disclosed. Although companies tend to underreport environmental data, the level of environmental decoupling decreased in 2023, demonstrating that the introduction of more stringent requirements for environmental disclosure (i.e., the Corporate Sustainability Reporting Directive 2022/2464/EU) could promote better alignment between sustainability disclosure and performance. The findings provide important recommendations for companies, regulators, and standard setters on how to design and configure the board of directors to align environmental disclosure and performance.

## 1 | Introduction

In recent decades, corporations have been paying increasing attention to corporate social responsibility (CSR) issues due to increasing pressure from stakeholders. As a result, companies are increasingly implementing actions aimed at achieving sustainable development and are enhancing the sustainability information released to external stakeholders (Gull, Hussain, Khan, Nadeem, and Zalata 2023; Porter and Kramer 2011) to obtain legitimacy and consensus on their activities (Hussain et al. 2023; Khan et al. 2015).

The increasing release of sustainability information is also a consequence of institutional pressure. In recent years,

regulators have introduced different rules aimed at encouraging companies to disclose sustainability data (European Parliament and the Council 2014). In 2014, the European Union (EU) issued the Non-Financial Reporting Directive 2014/95/EU (“Directive 2014/95/EU” hereafter), which introduced the requirement for bigger companies to disclose data on sustainability issues within a specific document, namely, the nonfinancial statement. Bigger European companies had to prepare a nonfinancial statement for the first time in 2018; this referred to the 2017 financial year. In 2022, the EU issued the Corporate Sustainability Reporting Directive 2022/2464/EU with the aim of further increasing the amount of sustainability information companies had to report (European Commission 2022).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2025 The Author(s). *Business Strategy and the Environment* published by ERP Environment and John Wiley & Sons Ltd.

Although companies are required to disclose enhanced sustainability information to external stakeholders, it is possible that companies do not report all the sustainability activities carried out (García-Sánchez et al. 2022; Sauerwald and Su 2019; Shahab et al. 2022; Tashman et al. 2019). This phenomenon is called the “CSR gap” or “decoupling” and refers to the difference between what companies disclose regarding their sustainability activities and what they really do (Tashman et al. 2019).

The existence of CSR decoupling generates doubt in stakeholders, which could reduce the perceived reliability and credibility of the sustainability information released (Boiral et al. 2019). Consequently, the presence of CSR misalignment between sustainability disclosure and performance could damage the company's reputation (Du 2015), decrease the value of the firm (Hawn and Ioannou 2016), and limit the company's capacity to access finance (García-Sánchez et al. 2021), thus resulting in increased agency problems between the company and its stakeholders (García-Sánchez et al. 2021). CSR decoupling reflects the information asymmetry between managers and other stakeholders as managers may try to show a positive environmental image through sustainability reports without implementing real actions. Agency theory explains how a strong board of directors can reduce information asymmetry and mitigate agency problems by monitoring management and ensuring accountability (Jensen and Meckling 1976).

Previous studies on CSR decoupling have revealed a lack of research on the determinants of this gap (Graafland and Smid 2019; Sauerwald and Su 2019). Few studies have investigated how different corporate governance characteristics, such as board gender diversity (Gull, Hussain, Khan, Nadeem, and Zalata 2023), the presence of a CSR committee (Gull, Hussain, Khan, Khan, and Saeed 2023), and CEO power (Gull, Hussain, Khan, Mushtaq, and Orij 2023; Shahab et al. 2022), influence CSR decoupling. Extant research has mainly focused on CSR decoupling and conducted longitudinal analysis on US or international samples, investigating the period before 2017. However, numerous scholars (Gull, Hussain, Khan, Khan, and Saeed 2023; Hussain et al. 2018) have highlighted that it is vitally important to analyze the decoupling of different sustainability dimensions, such as the environmental or social dimensions, instead of investigating the wider CSR gap. In addition, to the best of our knowledge, no studies have yet investigated the presence/absence of this gap following the introduction of more stringent requirements regarding the release of sustainability information. We select board independence, gender diversity, and the existence of a CSR committee to better reflect on the focus of Directive 2014/95/EU, which mandates accountability and transparency in nonfinancial disclosure within EU companies. For example, Article 18 of the directive requires companies to disclose corporate board diversity policies reflecting on the EU's recognition of board gender diversity as an important governance mechanism. In addition, the directive requires firms to report on environmental practices that are expected to be stronger with the presence of a dedicated CSR committee that oversees corporate sustainability strategy and reporting. Finally, the directive demands of disclosure related to risk management policies assert the need for an independent board that can effectively monitor management practices including sustainability reporting.

Therefore, besides the theoretical background, these factors better match with the regulatory expectations in the EU post-implementation period.

This paper makes several contributions to literature. First, this study examines a specific sustainability dimension, namely, environmental sustainability, by analyzing how various board characteristics impact environmental decoupling during mandatory sustainability disclosure periods. Second, this research fills an existing gap by focusing on the European context and investigating the influence of governance following the implementation of Directive 2014/95/EU. This directive mandates comprehensive sustainability reporting and is therefore a crucial setting for examining the impact of corporate governance on environmental outcomes. Third, this study concentrates on three critical board characteristics, namely, board independence, gender diversity, and the presence of a CSR committee. Previous studies used these mechanisms to investigate their capacity to influence either environmental disclosure or environmental performance (e.g., Agarwala et al. 2023; Gerged et al. 2021; Injeni et al. 2021; Orazalin 2019). However, less is known about how these board characteristics affect environmental decoupling or the gap between disclosure and actions. Therefore, we investigate these mechanisms to deepen the understanding of the board characteristics role in environmental misalignment. More specifically, the mechanisms identified are scrutinized regarding the role they play in influencing corporate environmental decisions. Within the agency theory framework (Jensen and Meckling 1976), these mechanisms serve to curb managerial opportunism and aid in aligning managers' actions with stakeholders' interests. Finally, this study provides evidence that effective corporate governance, as reflected by the aforementioned board characteristics, generally enhances the transparency and quality of the information disclosed to stakeholders as improved transparency is crucial in reducing the environmental performance gap (Sauerwald and Su 2019). By examining these aspects, the current study not only contributes to a deeper understanding of the mechanisms that mitigate agency problems; it also provides practical insights for practitioners and policymakers who aim to improve corporate environmental performance through strong corporate governance.

We conducted an empirical analysis on a sample of 728 European companies (3061 firm-year observations) belonging to 18 industries and 20 different countries between 2017 and 2023. To test the influence of the identified corporate governance characteristics on environmental decoupling, we employed fixed-effects estimations for panel data followed by two-stage least squares (2SLS) to address any possible endogeneity. The results show that companies tend to not disclose all the environmental actions implemented; they thus adopt underreporting behavior. The results also reveal that board independence, board gender diversity, and the presence of a CSR committee foster a reduction in environmental decoupling.

The remainder of this paper is structured as follows. Section 2 analyzes previous studies useful in developing our hypotheses. Section 3 outlines the methodology, whereas Section 4 presents the results of the study. Section 5 discusses our results and concludes the paper by outlining limitations and avenues for future research.

## 2 | Literature Review and Research Hypotheses

### 2.1 | Environmental Decoupling

Environmental matters are attracting increasing attention worldwide, with issues such as climate change, global warming, and deforestation prompting companies to integrate sustainability principles into their strategy, business model, and processes (Engert et al. 2016; Geissdoerfer et al. 2018; Meuer et al. 2020). Therefore, companies started to be responsible for sustainability matters by aligning their business with the sustainability principles (Broman and Robert 2017) and implementing business models able to contribute to the achievement of the sustainable development (Manninen and Huiskonen 2022).

In addition to integrate sustainability within their strategies and processes, the increasing attention towards environmental issues prompted companies to disclose a greater amount of environmental information in response to stakeholder pressure (Roberts et al. 2022; Sinclair-Desgagné et al. 2003) and in order to obtain legitimacy and consensus (De Colle et al. 2014; Lepore et al. 2023; Lock and Seele 2016; Stewart and Niero 2018).

For many years, companies released environmental information on a largely voluntary basis. However, in recent years, companies have increased the amount of environmental information they release in response to institutional pressure resulting from the issuing of different rules that require companies to disclose data on environmental issues. With reference to the European context, the EU issued Directive 2014/95/EU in 2014; starting from the 2017 fiscal year, bigger companies have been required to publish, among other things, environmental information in a nonfinancial statement.

The necessity for companies to pay attention to environmental topics and release environmental information has been further emphasized by recent EU regulations (i.e., Regulation 2019/2088 and the “taxonomy” regulation, Regulation 2020/852). These aim at promoting sustainable growth by encouraging the allocation of private capital flows into green investments.

Despite the growing desire of the EU to make the disclosure of environmental information mandatory, companies still have room for maneuver when it comes to deciding what environmental data to disclose. In particular, although Directive 2014/95/EU requires bigger companies to release environmental data, it does not specify in detail the information to be disclosed. This means that companies are free to choose the specific items of information they wish to disclose, as well as the reporting standard to refer to when releasing this kind of data. Moreover, Directive 2014/95/EU does not apply to small and medium-sized companies; such firms can voluntarily decide whether to disclose environmental information and what kind of information they wish to disclose.

The existence of different degrees of freedom and subjectivity when defining the environmental information to release (Marquis and Qian 2014), coupled with the presence of information asymmetries between companies and their stakeholders, could foster the adoption of opportunistic managerial

behaviors—manifested in the provision of unreliable, incomplete, and inaccurate information with the aim of showing a biased picture of a company's performance (Arena et al. 2018; Du and Wu 2019; Tata and Prasad 2015). For example, companies could decide to exclusively report information on favorable elements of environmental performance while hiding any unfavorable aspects (García-Sánchez et al. 2022).

The literature defines the abovementioned phenomenon as environmental decoupling, and it occurs when a company releases information that is different from the reality of the company's actions, thus creating a misalignment between the released environmental disclosure and the company's environmental performance (Tashman et al. 2019). Most previous studies have identified environmental decoupling as a form of greenwashing (Sauerwald and Su 2019), which occurs when the gap is positive, that is, when the environmental disclosure exceeds environmental performance. Therefore, greenwashing—or positive decoupling—occurs when a company releases more information in comparison to the actions it undertakes, thereby overstating its performance. Decoupling can also assume negative values, such as when a company decides to provide less information in comparison to the actions it undertakes, thereby understating its performance. The negative gap occurs when the environmental disclosure is lower than the company's environmental performance.

Companies may decide to report inaccurate information about their activities for numerous reasons. For example, companies could decide to disclose more information, thus enhancing the reality of their actions, as a result of them being subjected to high levels of stakeholder pressure (Crilly et al. 2012). At the same time, companies could decide to release less information, thus downplaying the reality of their actions, because of uncertainty regarding market reaction to their disclosure (García-Sánchez et al. 2022).

Independent of the nature of the decoupling, whether positive or negative, misalignment between what is disclosed, and the actions of a company results in an opaque and incorrect image of the company's performance (Crilly et al. 2012), thus reducing the credibility and reliability of the environmental information released (Boiral et al. 2019). Consequently, the existence of an environmental gap may damage the company's reputation (Du 2015), decrease the value of the firm (Hawn and Ioannou 2016), and limit the company's capacity to access finance (García-Sánchez et al. 2021).

### 2.2 | Theoretical Framework

To study the influence of board characteristics on environmental decoupling the paper referred to the agency theory, according to which the release of reliable environmental disclosure is considered a control mechanism useful for decreasing information asymmetries between the agent and the principal.

According to the agency theory (Jensen and Meckling 1976), an agency relationship occurs when a principal assigns to an agent the management of some activities on his behalf. The separation between ownership and control (Fama and

Jensen 1983) produces an information asymmetry between the agent and the principal such that, in making his decisions, the agent can refer to more company private information than the principal.

However, the choices assumed by the agent produce effects on both its own interests and those of the principal. Therefore, in cases of divergence of interests, the agent could assume some decisions in his interest decreasing the principal's welfare. The adoption of this opportunistic behavior by the agent generates a conflict, better known as agency problem.

There are two main agency problems that can occur within the companies. The first type of agency problems occurs between shareholders and managers. The second type of agency problems, more descriptive of different European contexts, occurs between minority shareholders and majority shareholder/manager. In both these types of agency problems, the principal operates outside the company, whereas the agent works within the company.

In these situations, the adoption of CSR activities favors the alignment between both the agent and the principal interests (Saeed et al. 2025), reducing the agency problems. At the same time, the release of environmental information also allows the decrease of the agency problems (Barako et al. 2006), letting the principal to better monitor the behaviors of the agent. Therefore, the environmental disclosure is a control mechanism useful to reduce the information asymmetries between the agent and the principal.

However, the presence of information asymmetries between the agent and the principal could bring the former in engaging in environmental decoupling, decreasing the credibility of the information disclosed (Boiral et al. 2019) and enhancing the agency problems (Saeed et al. 2025). Therefore, companies should avoid the environmental decoupling to reduce the agency problems.

In this context, the presence of an effective board of directors, ensuring the effective monitoring of the opportunistic behaviors of the agent and promoting the company transparency (Alfraih 2016), could facilitate the alignment between the environmental disclosure released and the environmental performance, reducing the environmental decoupling. Among the main board characteristics analyzed, this paper specifically focused on board independence, gender diversity, and the presence of a CSR committee.

## 2.3 | Hypothesis Development

### 2.3.1 | Board Independence and Environmental Decoupling

Board independence is considered an effective corporate governance mechanism when it comes to promoting environmental disclosure. According to agency theory, the presence of independent directors on the board ensures the effective monitoring of opportunistic managerial behaviors (Fama 1980; Hillman and Dalziel 2003); this increases the amount of information

released to external stakeholders (Garcia-Meca and Sanchez-Ballesta 2010) and consequently reduces agency problems.

The effectiveness of independent directors in encouraging environmental disclosure within companies has been investigated also using other theoretical approaches. According to the stakeholder theory framework, independent directors are expected to promote a higher level of environmental disclosure because they are more disposed to satisfy all stakeholders' information needs (Hussain et al. 2018) due to the lower pressure they experience from majority shareholders and managers compared to other board members. Moreover, independent directors tend to operate in the interest of a larger number of stakeholders (Armstrong et al. 2010; Prado-Lorenzo and Garcia-Sanchez 2010) because of the higher reputational concerns they are subject to. In this sense, independent directors are more inclined to release higher levels of environmental disclosure to obtain and maintain their professional reputation (Muttakin and Subramaniam 2015).

Most previous empirical studies have hypothesized—and found—a positive relationship between board independence and environmental disclosure (e.g., Gerged et al. 2021; Katmon et al. 2019; Pucheta-Martínez and López-Zamora 2018; Zaid et al. 2020). In addition, most previous research on the association between independent directors and environmental performance has also found a positive and statistically significant relationship between these two variables (e.g., Agarwala et al. 2023).

With specific reference to CSR decoupling, few studies have investigated the relationship between board independence and CSR decoupling (Gull, Hussain, Khan, Khan, and Saeed 2023; Gull, Hussain, Khan, Mushtaq, and Orij 2023; Gull, Hussain, Khan, Nadeem, and Zalata 2023). Most consider the presence of independent directors as a control variable and find a negative association between board independence and the CSR gap.

Based on previous theoretical arguments, this paper views board independence as a corporate governance mechanism capable of reducing environmental decoupling. We therefore state the following hypothesis:

**Hypothesis 1.** *There is a negative association between board independence and environmental decoupling.*

### 2.3.2 | Board Gender Diversity and Environmental Decoupling

In recent years, there has been a growing focus on board gender diversity, as achieving gender equality is one of the main Sustainable Development Goals (SDGs) of Agenda 2030. Consequently, several countries have enacted laws and corporate governance codes aimed at increasing the presence of women on boards. Women in management roles are better suited to supervisory activities, such as advising and monitoring functions (e.g., Adams and Ferreira 2009; Gul et al. 2008), and they thus facilitate a reduction in opportunistic managerial behaviors (Beetsma et al. 2000; Fama and Jensen 1983). A more gender-diverse board structure should reduce information asymmetries between a company and its stakeholders, as

women—due to their characteristics of being more supportive, respectful, kind, and empathetic than men—pay special attention to and try to meet the information needs of stakeholders (Katmon et al. 2019). Therefore, the presence of women on boards should promote the release of higher levels of environmental disclosure, given the fact that women are more stakeholder oriented (Injeni et al. 2021) and have a greater focus on sustainability issues (Hussain et al. 2018; Matsa and Miller 2013).

Extant literature has investigated the association between board gender diversity and the wider concept of CSR disclosure, mainly revealing that the presence of women directors on the board increases the level of information released (García-Sánchez et al. 2014; Hussain et al. 2018; Injeni et al. 2021; Liao et al. 2015). Most studies have also found a positive association between board gender diversity and CSR performance (Atif et al. 2020; Post et al. 2011).

With respect to decoupling, few studies have investigated the relationship between board gender diversity and CSR decoupling (García-Sánchez et al. 2022; Gull, Hussain, Khan, Nadeem, and Zalata 2023; Sauerwald and Su 2019; Shahab et al. 2022), and they have mainly considered the presence of women on the board as a control variable. Interestingly, these studies have reported contrasting results. Gull, Hussain, Khan, Nadeem, and Zalata (2023) found a negative and significant relationship between gender diversity and environmental misalignment. However, other studies (i.e., García-Sánchez et al. 2022; Gull, Hussain, Khan, Mushtaq, and Orij 2023) found a positive and statistically significant association between women on the board and CSR decoupling, with these findings running contrary to theoretical arguments.

Based on previous theoretical arguments, this paper considers the presence of women on boards as a corporate governance mechanism capable of reducing environmental decoupling. We therefore state the following hypothesis:

**Hypothesis 2.** *There is a negative association between board gender diversity and environmental decoupling.*

### 2.3.3 | CSR Committees and Environmental Decoupling

Companies frequently establish a CSR committee within the board of directors with the aim of managing CSR issues. The CSR committee is mainly composed of directors with different experience and knowledge of CSR-related matters. The literature suggests that the establishment of a CSR committee is a signal of a company being orientated toward and paying attention to CSR issues (Gennari and Salvioni 2019); furthermore, it ensures the adoption of CSR strategies (Klettner et al. 2014) and improves the company's CSR engagement (Hussain et al. 2018). According to the agency theory framework, the existence of a CSR committee fosters better monitoring of opportunistic managerial behaviors, thus reducing agency problems between the company and its stakeholders. Previous studies on the effects of the existence of a CSR committee support the existence of a positive relationship with both environmental disclosure (Giannarakis et al. 2020; Helfaya

and Moussa 2017; Peters and Romi 2014) and environmental performance (Godos-Díez et al. 2018; Govindan et al. 2021; Orazalin 2019).

With respect to decoupling between disclosure and performance, the study of Gull, Hussain, Khan, Khan, and Saeed (2023) specifically investigated the relationship between the presence of a CSR committee and the CSR gap, finding that the existence of a CSR committee within the board is significantly and negatively associated with CSR decoupling. The study further revealed that the presence of a CSR committee is an effective corporate governance mechanism that reduces the misalignment between disclosure and performance.

Based on previous theoretical arguments, this paper considers the presence of a CSR committee as a corporate governance mechanism capable of reducing environmental decoupling. We therefore state the following hypothesis:

**Hypothesis 3.** *There is a negative association between the presence of a CSR committee and environmental decoupling.*

## 3 | Research Methods

### 3.1 | Sample Selection and Data Collection

We conducted an analysis of a sample of European listed companies. We selected all companies whose country of both headquarters and exchange is a European member state. Among these companies, we chose those that had prepared a sustainability report. We collected data from the Refinitiv Eikon database for the period from 2017 to 2023. We decided to start the analysis in 2017, as this was the first year in which Directive 2014/95/UE was implemented. Our population comprised 1225 European listed companies (9800 firm-year observations).

We excluded from the population all companies that did not share environmental, social, and governance (ESG) data, information on board composition, and financial data useful for our analysis on the Refinitiv Eikon database. The final unbalanced sample was composed of 728 companies (3061 firm-year observations) belonging to 18 industries and 20 different countries.

Table 1 shows the sample distribution by country of headquarters (Panel A), industry (Panel B), and year (Panel C).

Panel A reports the distribution of companies by country headquarters. Most companies are from Germany (19.76%), followed by Sweden (18.65%), France (14.99%), Italy (10.98%), Spain (7.55%), Finland (7.32%), and the Netherlands (6.30%). The other countries present a percentage of observations lower than 5%.

Panel B of Table 1 describes the industries in which the sampled companies operate. Most companies belong to the manufacturing sector (44.95%), followed by information (8.23%), real estate and rental and leasing (6.96%), utilities (5.91%), finance and insurance (5.49%), construction (5.16%), and professional,

**TABLE 1** | Sample distribution by country of headquarters, industry, and year.

| <b>Panel A. Sample by country of headquarters</b>  |                     |                    |                  |                                                  |                     |                    |               |
|----------------------------------------------------|---------------------|--------------------|------------------|--------------------------------------------------|---------------------|--------------------|---------------|
| <b>Country</b>                                     | <b>No. of firms</b> | <b>No. of obs.</b> | <b>% obs.</b>    | <b>Country</b>                                   | <b>No. of firms</b> | <b>No. of obs.</b> | <b>% obs.</b> |
| Austria                                            | 22                  | 95                 | 3.10%            | Luxembourg                                       | 4                   | 17                 | 0.56%         |
| Belgium                                            | 30                  | 105                | 3.43%            | Malta                                            | 1                   | 6                  | 0.20%         |
| Czech Republic                                     | 1                   | 5                  | 0.16%            | Netherlands                                      | 38                  | 193                | 6.30%         |
| Denmark                                            | 20                  | 68                 | 2.22%            | Poland                                           | 7                   | 17                 | 0.56%         |
| Finland                                            | 47                  | 224                | 7.32%            | Portugal                                         | 13                  | 68                 | 2.22%         |
| France                                             | 93                  | 459                | 14.99%           | Romania                                          | 1                   | 2                  | 0.07%         |
| Germany                                            | 140                 | 605                | 19.76%           | Slovenia                                         | 1                   | 2                  | 0.07%         |
| Greece                                             | 12                  | 32                 | 1.04%            | Spain                                            | 58                  | 231                | 7.55%         |
| Hungary                                            | 2                   | 8                  | 0.26%            | Sweden                                           | 139                 | 571                | 18.65%        |
| Ireland                                            | 6                   | 17                 | 0.56%            | <i>All countries</i>                             | <i>728</i>          | <i>3061</i>        | <i>100%</i>   |
| Italy                                              | 93                  | 336                | 10.98%           |                                                  |                     |                    |               |
| <b>Panel B. Sample by industry</b>                 |                     |                    |                  |                                                  |                     |                    |               |
| <b>Industry</b>                                    | <b>No. of firms</b> | <b>No. of obs.</b> | <b>% obs.</b>    | <b>Industry</b>                                  | <b>No. of firms</b> | <b>No. of obs.</b> | <b>% obs.</b> |
| Accommodation and food services                    | 7                   | 35                 | 1.14%            | Mining, quarrying, oil, and gas extraction       | 11                  | 40                 | 1.31%         |
| Admin. and supp., waste manag. and remed. services | 17                  | 63                 | 2.06%            | Other services (except public administration)    | 3                   | 13                 | 0.42%         |
| Agricult., forestry, fishing and hunting           | 2                   | 5                  | 0.17%            | Professional, scientific, and technical services | 40                  | 154                | 5.03%         |
| Arts, entertainment, and recreation                | 5                   | 23                 | 0.75%            | Real estate and rental and leasing               | 55                  | 213                | 6.96%         |
| Construction                                       | 40                  | 158                | 5.16%            | Retail trade                                     | 33                  | 145                | 4.74%         |
| Educational services                               | 1                   | 5                  | 0.17%            | Transportation and warehousing                   | 30                  | 135                | 4.41%         |
| Finance and insurance                              | 43                  | 168                | 5.49%            | Utilities                                        | 40                  | 181                | 5.91%         |
| Health care and social assistance                  | 8                   | 22                 | 0.72%            | Wholesale trade                                  | 17                  | 73                 | 2.38%         |
| Information                                        | 62                  | 252                | 8.23%            | <i>All industries</i>                            | <i>728</i>          | <i>3061</i>        | <i>100%</i>   |
| Manufacturing                                      | 314                 | 1376               | 44.95%           |                                                  |                     |                    |               |
| <b>Panel C. Sample by year</b>                     |                     |                    |                  |                                                  |                     |                    |               |
| <b>Year</b>                                        | <b>No. of obs.</b>  | <b>% obs.</b>      | <b>Year</b>      | <b>No. of obs.</b>                               | <b>% obs.</b>       |                    |               |
| 2017                                               | 212                 | 6.93%              | 2021             | 516                                              | 16.86%              |                    |               |
| 2018                                               | 270                 | 8.82%              | 2022             | 604                                              | 19.73%              |                    |               |
| 2019                                               | 381                 | 12.45%             | 2023             | 651                                              | 21.26%              |                    |               |
| 2020                                               | 427                 | 13.95%             | <i>All years</i> | <i>3061</i>                                      | <i>100%</i>         |                    |               |

scientific, and technical services (5.03%). The other industries present a percentage of observations lower than 5%.

Finally, Panel C of Table 1 shows the distribution of the observations by year. The higher concentration of the observation number was consolidated in the years between 2021 and 2023.

The lower concentration of the number of observations was consolidated in the years between 2017 and 2018. The years 2019 and 2020 begin to show an increase in the distribution of the observations by year. The number of observations has risen over the years, revealing an increase in the availability of data on the Refinitiv Eikon database concerning the variables investigated.

### 3.2 | Dependent, Independent, and Control Variables

We followed previous highly regarded studies to measure environmental decoupling. (García-Sánchez et al. 2022; Gull, Hussain, Khan, Khan, and Saeed 2023; Hawn and Ioannou 2016). This proxy is widely used in the literature and works well with panel data models. This also has enabled us to better compare our findings with those of earlier studies on CSR decoupling while placing emphasis on environmental issues. More specifically, this study measures the dependent variable—that is, environmental decoupling (*EnvDecoup*)—as the difference between the environmental disclosure (*EnvDisc*) released by a company and its environmental performance (*EnvPerf*). To compute *EnvDisc*, we referred to the Refinitiv Eikon database and selected all the environmental items requiring companies to “report,” “claim,” or “disclose” environmental information. We identified 36 environmental items, as reported in Appendix 1. For each environmental item selected, Refinitiv assigns a value of 1 when the company discloses information on that environmental aspect and 0 otherwise. *EnvDisc* was measured as the unweighted sum of the 36 environmental items. Following previous studies (García-Sánchez et al. 2022; Gull, Hussain, Khan, Khan, and Saeed 2023; Hawn and Ioannou 2016), we normalized the *EnvDisc* variable, obtaining a 0–1 scale. Therefore, we divided the environmental items disclosed by each company by the total number of environmental aspects that companies should disclose as follows:

$$\text{EnvDisc}_{it} = \frac{\sum \text{Environmental Items}_{it}}{36}$$

Therefore, the *EnvDisc* released by company *i* in year *t* is equal to the sum of environmental items disclosed divided by the 36 environmental aspects identified. To measure *EnvPerf*, we also referred to the Refinitiv Eikon database and selected all the environmental items describing the environmental actions adopted by companies, in terms of both the policies implemented and the targets defined by companies. We identified 37 environmental items, as reported in Appendix 1. For each environmental item selected, Refinitiv assigns a value of 1 when the company adopts the policy or defines a target on that environmental aspect and 0 otherwise. *EnvPerf* was measured as the unweighted sum of the 37 environmental items. Similar to the *EnvDisc* variable, we normalized the *EnvPerf* variable, obtaining a 0–1 scale. Therefore, we divided the environmental items computed for each company by the total number of environmental aspects identified as follows:

$$\text{EnvPerf}_{it} = \frac{\sum \text{Environmental Items}_{it}}{37}$$

Therefore, the *EnvPerf* assigned to company *i* in year *t* is equal to the sum of environmental items computed for each company divided by the 37 environmental aspects identified.

To compute *EnvDecoup*, we made the differences between the *EnvDisc* released in year *t* and the *EnvPerf* of the year *t* – 1, lagging the *EnvPerf* component as follows:

$$\text{EnvDecoup}_i = \text{EnvDisc}_{it} - \text{EnvPerf}_{it-1}$$

Higher values of *EnvDecoup* show the presence of a misalignment between the environmental information released and environmental performance. This misalignment may be both positive (i.e., *EnvDisc* is higher than *EnvPerf*) and negative (i.e., *EnvDisc* is lower than *EnvPerf*). The *EnvDecoup* variable can assume values ranging from –1 (in the case of companies that do not disclose all their environmental actions, namely underreporting companies) to 1 (in the case of companies that disclose more actions than they undertake in reality, viz., underperforming companies).

The independent variables are board gender diversity (*BoGenDivers*), measured as the percentage of women directors sitting on the board; board independence (*IndepBoMemb*), computed as the percentage of independent board members; and CSR committee (*CSRCom*), measured using a dichotomous variable equal to 1 if the company has a CSR committee and 0 otherwise. The decision to focus on these board characteristics derived from the consideration that less is known about their capacity to affect environmental decoupling, that is to say, the combined environmental disclosure and performance.

As suggested by previous research (García-Sánchez et al. 2022; Gull, Hussain, Khan, Khan, and Saeed 2023), we inserted different control variables into our analysis that could impact the level of *EnvDecoup*. These variables are board size (*BoSize*), measured as the total number of board members; role duality (*CEOChairDual*), measured using a dummy variable equal to 1 if the company has a CEO who is also the chairman of the board and 0 otherwise; board meeting (*NBoMeet*), measured as the number of meetings during the year; audit board committee (*AudBoCom*), computed using a dummy variable equal to 1 if the company has an audit board committee and 0 otherwise; board-specific skills (*BoSpecSkill*), measured as the percentage of board members who have either an industry-specific background or a strong financial background; average board tenure (*AveBoTen*), computed as the average number of years each board member has been on the board; non-executive board members (*NonExecBoMemb*), computed as the percentage of non-executive board members; stakeholder engagement (*StakeEngag*), measured using a dummy variable equal to 1 if the company explains how it engages with its stakeholders and 0 otherwise; external assurance (*CSRStustExternAud*), measured using a dummy variable equal to 1 if the company requires the assurance of its sustainability report and 0 otherwise; firm size (*size*), measured as the natural logarithm of total assets; and leverage (*Lev*), computed as total long-term debts divided by total assets.

Table 2 summarizes all the variables included in the analysis and describes how they have been measured; it also lists the data sources.

### 3.3 | Regression Analysis

To assess the effect of *IndepBoMemb*, *BoGenDivers*, and *CSRCom* on *EnvDecoup*, we developed the following linear regression model:

**TABLE 2** | Variable description, measurement, and source.

| Variable                | Description                 | Measurement                                                                                                                                                                                                                                                                                                                         | Source |
|-------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Dependent variable      |                             |                                                                                                                                                                                                                                                                                                                                     |        |
| <i>EnvDecoup</i>        | Environmental decoupling    | Difference between 36 environmental disclosure ( <i>EnvDisc</i> ) items released by company and 37 environmental performance ( <i>EnvPerf</i> ) items. Both <i>EnvDisc</i> and <i>EnvPerf</i> were normalized by dividing the environmental items computed for each company by the total number of environmental aspects identified | Eikon  |
| Independent variables   |                             |                                                                                                                                                                                                                                                                                                                                     |        |
| <i>IndepBoMemb</i>      | Board independence          | Percentage of independent members sitting on the board                                                                                                                                                                                                                                                                              | Eikon  |
| <i>BoGenDivers</i>      | Board gender diversity      | Percentage of women directors sitting on the board                                                                                                                                                                                                                                                                                  | Eikon  |
| <i>CSRCom</i>           | CSR committee               | Dichotomous variable equal to 1 if the company has a CSR committee and 0 otherwise                                                                                                                                                                                                                                                  | Eikon  |
| Control variables       |                             |                                                                                                                                                                                                                                                                                                                                     |        |
| <i>BoSize</i>           | Board size                  | Total number of board members at the end of the fiscal year                                                                                                                                                                                                                                                                         | Eikon  |
| <i>CEOChairDual</i>     | CEO chairman duality        | Dummy variable equal to 1 if the company has the CEO that is also the chairman of the board and 0 otherwise                                                                                                                                                                                                                         | Eikon  |
| <i>NBoMeet</i>          | Board meetings              | Number of board meetings during the year                                                                                                                                                                                                                                                                                            | Eikon  |
| <i>AudBoCom</i>         | Audit board committee       | Dummy variable equal to 1 if the company has an audit board committee and 0 otherwise                                                                                                                                                                                                                                               | Eikon  |
| <i>BoSpecSkill</i>      | Board specific skills       | Percentage of board members who have either an industry specific background or a strong financial background                                                                                                                                                                                                                        | Eikon  |
| <i>AveBoTen</i>         | Average board tenure        | Average number of years each board member has been on the board                                                                                                                                                                                                                                                                     | Eikon  |
| <i>NonExecBoMemb</i>    | Non-executive board members | Percentage of non-executive board members                                                                                                                                                                                                                                                                                           | Eikon  |
| <i>StakeEngag</i>       | Stakeholder engagement      | Dummy variable equal to 1 if the company explains how it engages with its stakeholders and 0 otherwise                                                                                                                                                                                                                              | Eikon  |
| <i>CSRSustExternAud</i> | External assurance          | Dummy variable equal to 1 if the company requires the assurance of its sustainability report and 0 otherwise                                                                                                                                                                                                                        | Eikon  |
| <i>size</i>             | Firm size                   | Natural logarithm of total assets                                                                                                                                                                                                                                                                                                   | Eikon  |
| <i>Lev</i>              | Leverage                    | Long-term debt divided by total assets                                                                                                                                                                                                                                                                                              | Eikon  |

$$\begin{aligned}
 \text{EnvDecoup} = & \alpha + \beta_1 \text{IndepBoMemb} + \beta_2 \text{BoGenDivers} \\
 & + \beta_3 \text{CSRCom} + \beta_4 \text{BoSize} + \beta_5 \text{CEOChairDual} \\
 & + \beta_6 \text{NBoMeet} + \beta_7 \text{AudBoCom} + \beta_8 \text{BoSpecSkill} \\
 & + \beta_9 \text{AveBoTen} + \beta_{10} \text{NonExecBoMemb} + \beta_{11} \text{CSRSustExternAud} \\
 & + \beta_{12} \text{StakeEngag} + \beta_{13} \text{size} + \beta_{14} \text{Lev} + \varepsilon
 \end{aligned}$$

*EnvDecoup* presents a mean value of  $-0.0635$ , revealing the presence of a misalignment between *EnvDisc* and *EnvPerf*. More specifically, the negative average value of *EnvDecoup* indicates that companies tend to not disclose all the environmental actions implemented. This finding is consistent with the results of Gull, Hussain, Khan, Khan, and Saeed (2023), Gull, Hussain, Khan, Nadeem, and Zalata (2023), who found a negative gap in a sample of listed companies from 41 different countries in the period from 2003 to 2017 and in a sample of US firms from 2002 to 2017, respectively. The minimum value is  $-0.375$ , and the maximum value is  $0.417$ . These data show that this gap is negative for some sampled companies, revealing that they are underreporting companies; for others, however, the misalignment is positive, revealing that they are underperforming companies.

## 4 | Results

### 4.1 | Descriptive Analysis

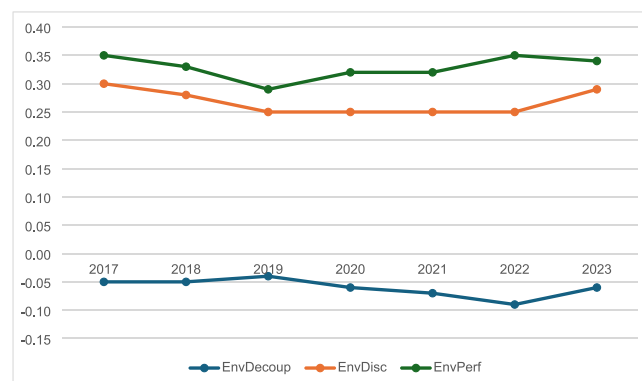
Table 3 reports the descriptive statistics for all the variables included in the analysis.

**TABLE 3** | Descriptive statistics.

| Variables               | (1)<br>N | (2)<br>Mean | (3)<br>SD | (4)<br>Min | (5)<br>Max |
|-------------------------|----------|-------------|-----------|------------|------------|
| Dependent variable      |          |             |           |            |            |
| <i>EnvDecoup</i>        | 3061     | −0.0635     | 0.113     | −0.375     | 0.417      |
| <i>EnvDisc</i>          | 3061     | 0.265       | 0.126     | 0          | 0.694      |
| <i>EnvPerf</i>          | 3061     | 0.363       | 0.117     | 0          | 0.703      |
| Independent variables   |          |             |           |            |            |
| <i>IndepBoMemb</i>      | 3061     | 59.65       | 25.38     | 0          | 100        |
| <i>BoGenDivers</i>      | 3061     | 34.20       | 11.73     | 0          | 75         |
| <i>CSRCom</i>           | 3061     | 0.851       | 0.356     | 0          | 1          |
| Control variables       |          |             |           |            |            |
| <i>BoSize</i>           | 3061     | 11.01       | 4.112     | 2          | 28         |
| <i>CEOChairDual</i>     | 3061     | 0.254       | 0.435     | 0          | 1          |
| <i>NBoMeet</i>          | 3061     | 10.64       | 6.021     | 2          | 122        |
| <i>AudBoCom</i>         | 3061     | 0.967       | 0.180     | 0          | 1          |
| <i>BoSpecSkill</i>      | 3061     | 28.96       | 17.10     | 0          | 100        |
| <i>AveBoTen</i>         | 3061     | 6.512       | 2.756     | 0.0978     | 23.17      |
| <i>NoExecBoMemb</i>     | 3061     | 88.63       | 12.48     | 33.33      | 100        |
| <i>StakEngag</i>        | 3061     | 0.925       | 0.263     | 0          | 1          |
| <i>CSRSustExternAud</i> | 3061     | 0.884       | 0.320     | 0          | 1          |
| <i>size</i>             | 3061     | 22.49       | 1.680     | 17.77      | 27.76      |
| <i>Lev</i>              | 3061     | 0.218       | 0.137     | 0          | 0.959      |

With respect to the components of *EnvDecoup*, *EnvDisc* presents a mean value of 0.265, revealing a low level of *EnvDisc*. The minimum value is 0, and the maximum value is 0.694. These data show that there are companies that provide no environmental information at all; at the same time, no company provided information on all the 36 environmental aspects identified. The *EnvPerf* variable presents similar results. The average value is 0.363, the minimum value is 0, and the maximum value is 0.703. Therefore, the sampled companies present low levels of *EnvPerf*. In addition, there are companies that did not adopt policies or define targets on environmental issues; furthermore, no company adopted environmental actions for all the 37 environmental aspects identified.

Figure 1 reports the trend of the mean values of the *EnvDecoup*, *EnvDisc*, and *EnvPerf* variables during the period investigated. The figure shows that after 2019, the year in which both *EnvDisc* and *EnvPerf* achieved their lowest levels, the sampled companies increased the number of policies adopted; moreover, the targets defined regarding environmental issues were higher in number than the information released on these topics. In 2023, *EnvDisc* increased for the sampled companies, probably in response to the Corporate Sustainability Reporting Directive 2022/2464/EU issued in December 2022 and the European Sustainability Reporting Standards issued by the European Financial Reporting Advisory Group (EFRAG) and adopted by

**FIGURE 1** | Trend of *EnvDecoup*, *EnvDisc*, and *EnvPerf*.

the European Commission in July 2023. Due to the increase in *EnvDisc*, the mean value of the *EnvDecoup* variable decreased in 2023, reducing from −0.09 in 2022 to −0.06 in 2023.

Moving on to board characteristics, the average number of board members is 11. More than 59% of board members, on average, are independent; more than 34% of board members, on average, are women; and more than 88% of board members, on average, are non-executive directors. Eighty-five percent of the sampled companies have a CSR committee, 96% of the companies have an audit board committee, and in 25% of the sampled companies,

the CEO is also the chairman of the board. Ninety-two percent of companies have a stakeholder engagement to better understand which sustainability issues are the most material to them. On average, 28% of board members have either an industry-specific background or a strong financial background. Indeed, the *BoSpecSkill* variable assumes a value from 0 to 100. In addition, the average number of board meetings held during the year is 10, and the average number of years each board member has been on the board is 6.

With respect to the independent variables, Figure 2 shows that the percentage of independent directors, as well as the proportion of women members on the board, did not substantially change during the years investigated. The percentage of companies with a CSR committee, however, increased after 2019 and remained largely unchanged in subsequent years.

With respect to the control variables, 92% of the companies explain how they engage with their stakeholders. Eighty-eight percent of the sampled companies require the assurance of their sustainability report, revealing the increasing attention paid to the credibility of the sustainability information released by companies. Finally, the companies are followed by 12 analysts on average.

Before performing the regression analysis, we analyzed the correlations between all the model variables (Table 4). *EnvDecoup* is negatively and significantly correlated with *EnvPerf*, *IndepBoMemb*, *BoGenDivers*, *CSRCom*, *NoExecBoMemb*, *StakeEngag*, *lev*, and *size*. At the same time, *EnvDecoup* is positively and significantly correlated with *EnvDisc* and *CEOChairDual*.

To control for multicollinearity, we first estimated a pooled OLS regression model (Menard 2002). Table 4 reports the variance inflation factor (VIF) for all the independent variables. The VIFs for all the variables are lower than the threshold value of 10, revealing that no multicollinearity exists in the multiple regression models (Hair et al. 1998; Tashman et al. 2019).

## 4.2 | Regression Results

To achieve the objectives of the study, we used panel data that spanned the period 2017–2023 and employed fixed-effects

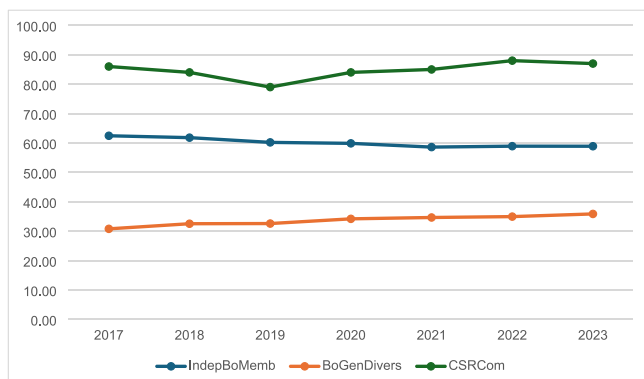


FIGURE 2 | Trend of *IndepBoMemb*, *BoGenDivers*, and *CSRCom*.

estimations. As the data have cross-sectional and time series dimensions, fixed-effects models control for unobserved heterogeneity that may vary across firms but remain constant over time. To ensure that our results were robust, we ran a series of regressions that included several control variables and year, industry, and country fixed effects, as this helps account for time-invariant characteristics within these categories that may influence environmental decoupling. This also ensured that our models accurately isolate the impact of our independent variables on environmental decoupling, leading to more reliable and robust results. We also used clustering at the firm level to control any cross-sectional dependence. In addition, as corporate governance research has been criticized of having endogeneity issues, we ran our main models in an additional analysis using 2SLS to control for any possible endogeneity. The regression results from Table 5 provide important insights into the factors influencing environmental decoupling (*EnvDecoup*), with a particular focus on independent board members (*IndepBoMemb*), board gender diversity (*BoGenDivers*), and the presence of a CSR committee (*CSRCom*).

As reported in the last column of Table 5, the higher the percentage of independent directors sitting on the board, the lower the environmental decoupling. The coefficient of *IndepBoMemb* is significantly and negatively related to *EnvDecoup* ( $\beta = -0.001$ ,  $p < 0.01$ ), supporting Hypothesis 1. This result is also confirmed across other models. This finding is consistent with the results of previous studies (Gull, Hussain, Khan, Khan, and Saeed 2023; Gull, Hussain, Khan, Mushtaq, and Orij 2023; Gull, Hussain, Khan, Nadeem, and Zalata 2023) revealing a negative association between board independence and the CSR gap. The negative association between board independence and environmental decoupling supports the positive effect that independent board members have on environmental disclosure, as revealed in previous studies (Fernández-Gago et al. 2018; Katmon et al. 2019; Zaid et al. 2020). The result is in line with the theoretical expectation and agency theory suggesting that independent directors improve the board monitoring functions and reduce information asymmetry between managers and stakeholders. This negative impact implies that firms with more independent board members are less likely to support symbolic environmental practices strategies and more likely to reduce the gap between environmental disclosure and performance.

Regarding Hypothesis 2, our findings reveal that *BoGenDivers* is negatively and weakly related to *EnvDecoup*. Therefore, the presence of women on the board weakly influences environmental decoupling, supporting the notion that board gender diversity plays an important role in shaping the CSR decisions made within a company. The significance of the coefficient does not hold in Models 3 and 4, casting doubt on the robustness of the results. The result suggests that board gender diversity may initially add value to the firm sustainable practices, but the impact disappears when controlling for industry and country effects. This result is partly consistent with the findings of Gull, Hussain, Khan, Nadeem, and Zalata (2023), who reported a negative relationship between gender diversity and environmental misalignment. However, this finding is inconsistent with those of García-Sánchez et al. (2022) and Gull, Hussain, Khan, Mushtaq, and Orij (2023), who found a positive and statistically significant association with CSR decoupling.

TABLE 4 | Correlation matrix.

|                  | Env<br>Decoup | EnvDisc  | EnvPerf  | Indep<br>BoMemb | BoGen<br>Divers | CSR<br>Com | BoSize   | CEO<br>Chair<br>Dual | NBo<br>Meet | AudBo<br>Com | BoSpec<br>Skill | Ave<br>BoTen | NoExec<br>BoMemb | Staka<br>Engag | CSR<br>Extern<br>Aud | size | Lev  | VIF  |
|------------------|---------------|----------|----------|-----------------|-----------------|------------|----------|----------------------|-------------|--------------|-----------------|--------------|------------------|----------------|----------------------|------|------|------|
| EnvDecoup        | 1.00          |          |          |                 |                 |            |          |                      |             |              |                 |              |                  |                |                      |      |      |      |
| EnvDisc          | 0.26***       | 1.00     |          |                 |                 |            |          |                      |             |              |                 |              |                  |                |                      |      |      | 2.40 |
| EnvPerf          | -0.26***      | 0.71***  | 1.00     |                 |                 |            |          |                      |             |              |                 |              |                  |                |                      |      |      | 2.38 |
| IndepBo<br>Memb  | -0.06***      | 0.00     | 0.01     | 1.00            |                 |            |          |                      |             |              |                 |              |                  |                |                      |      |      | 1.29 |
| BoGenDivers      | -0.08***      | 0.10***  | 0.13***  | 0.07***         | 1.00            |            |          |                      |             |              |                 |              |                  |                |                      |      |      | 1.09 |
| CSRCom           | -0.07***      | 0.33***  | 0.35***  | 0.03            | 0.09***         | 1.00       |          |                      |             |              |                 |              |                  |                |                      |      |      | 1.21 |
| BoSize           | -0.00         | 0.40***  | 0.36***  | -0.28***        | 0.03            | 0.20***    | 1.00     |                      |             |              |                 |              |                  |                |                      |      |      | 1.83 |
| CEOChair<br>Dual | 0.07***       | 0.19***  | 0.11***  | -0.15***        | 0.12***         | 0.09***    | 0.13***  | 1.00                 |             |              |                 |              |                  |                |                      |      |      | 1.18 |
| NBoMeet          | -0.01         | -0.08*** | -0.06*** | 0.12***         | -0.02           | 0.04       | -0.19*** | -0.03                | 1.00        |              |                 |              |                  |                |                      |      |      | 1.15 |
| AudBoCom         | -0.01         | 0.09***  | 0.08***  | 0.12***         | 0.07***         | 0.05*      | 0.08***  | 0.02                 | -0.01       | 1.00         |                 |              |                  |                |                      |      |      | 1.07 |
| BoSpecSkill      | -0.03         | -0.03    | -0.02    | -0.02           | 0.04*           | 0.02       | -0.18*** | 0.18***              | 0.06***     | -0.04*       | 1.00            |              |                  |                |                      |      |      | 1.35 |
| AveBoTen         | -0.03         | 0.02     | 0.01     | -0.14***        | -0.05***        | -0.05**    | 0.02     | 0.16***              | -0.17***    | -0.11***     | 0.12***         | 1.00         |                  |                |                      |      |      | 1.16 |
| NoExec<br>BoMemb | -0.06***      | 0.10***  | 0.13***  | 0.22***         | 0.02            | 0.07***    | 0.10***  | -0.23***             | -0.13***    | 0.07***      | -0.44***        | -0.24***     | 1.00             |                |                      |      |      | 1.50 |
| StakaEngag       | -0.08***      | 0.13***  | 0.18***  | 0.06**          | 0.01            | 0.16**     | 0.04*    | 0.00                 | 0.04*       | -0.01        | 0.02            | 0.01         | 0.00             | 1.00           |                      |      |      | 1.07 |
| CSR<br>ExternAud | -0.02         | 0.20***  | 0.19***  | -0.00           | 0.13***         | -0.04*     | 0.18***  | 0.05**               | -0.04*      | 0.14***      | 0.04*           | 0.02         | -0.04*           | 0.01           | 1.00                 |      |      | 1.11 |
| size             | -0.09***      | 0.52***  | 0.50***  | 0.07***         | 0.10***         | 0.25***    | 0.53***  | 0.10***              | -0.08***    | 0.16***      | -0.05***        | -0.03        | 0.19***          | 0.14***        | 0.20***              | 1.00 |      | 3.00 |
| Lev              | -0.03         | -0.06**  | -0.04*   | 0.02            | -0.00           | 0.06***    | -0.01    | -0.03                | 0.10***     | 0.02         | 0.04*           | -0.11***     | -0.03            | 0.06***        | 0.02                 | 0.03 | 1.00 | 1.11 |

\*\*\* $p < 0.001$ .\*\* $p < 0.01$ .\* $p < 0.05$ .

**TABLE 5** | Regression results (dependent variable is *EnvDecoup*).

| Variables               | Environmental decoupling |                       |                       |                       |
|-------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
|                         | Model 1                  | Model 2               | Model 3               | Model 4               |
| <i>IndepBoMemb</i>      | −0.001***<br>(−3.990)    | −0.001***<br>(−2.975) | −0.001***<br>(−3.659) | −0.001***<br>(−3.191) |
| <i>BoGenDivers</i>      | −0.001***<br>(−4.193)    | −0.001***<br>(−3.490) | −0.000<br>(−1.460)    | −0.000<br>(−1.250)    |
| <i>CSRCom</i>           | −0.052***<br>(−6.622)    | −0.039***<br>(−4.848) | −0.027***<br>(−3.341) | −0.027***<br>(−2.756) |
| <i>BoSize</i>           |                          | −0.001<br>(−0.472)    | −0.002<br>(−0.950)    | −0.002<br>(−0.771)    |
| <i>CEOChairDual</i>     |                          | −0.007<br>(−0.729)    | −0.009<br>(−0.943)    | −0.009<br>(−0.831)    |
| <i>NBoMeet</i>          |                          | 0.001<br>(1.216)      | 0.001**<br>(2.487)    | 0.001*<br>(1.887)     |
| <i>AudBoCom</i>         |                          | −0.005<br>(−0.310)    | −0.010<br>(−0.558)    | −0.010<br>(−0.393)    |
| <i>BoSpecSkill</i>      |                          | −0.000<br>(−0.216)    | −0.000<br>(−0.647)    | −0.000<br>(−0.541)    |
| <i>AveBoTen</i>         |                          | −0.007***<br>(−4.278) | −0.006***<br>(−3.881) | −0.006***<br>(−3.062) |
| <i>NoExecBoMemb</i>     |                          | −0.001*<br>(−1.667)   | 0.000<br>(0.208)      | 0.000<br>(0.152)      |
| <i>CSRSustExternAud</i> |                          | −0.024**<br>(−2.117)  | −0.017<br>(−1.517)    | −0.017<br>(−1.463)    |
| <i>StakeEngag</i>       |                          | −0.017<br>(−1.495)    | −0.006<br>(−0.558)    | −0.006<br>(−0.458)    |
| <i>Size</i>             |                          | −0.051***<br>(−5.635) | −0.013<br>(−1.292)    | −0.013<br>(−1.166)    |
| <i>Leverage</i>         |                          | −0.019<br>(−0.566)    | −0.025<br>(−0.754)    | −0.025<br>(−0.630)    |
| Constant                | 0.067***<br>(4.297)      | 1.346***<br>(6.765)   | 0.386*<br>(1.692)     | 0.386<br>(1.499)      |
| Observations            | 3061                     | 3061                  | 3061                  | 3061                  |
| <i>R</i> -squared       | 0.036                    | 0.066                 | 0.115                 | 0.115                 |
| Number of firms         | 728                      | 728                   | 728                   | 728                   |
| Control variables       | No                       | Yes                   | Yes                   | Yes                   |
| Year FE                 | No                       | No                    | Yes                   | Yes                   |
| Industry FE             | No                       | No                    | No                    | Yes                   |
| Country FE              | No                       | No                    | No                    | Yes                   |
| Cluster                 | No                       | No                    | No                    | Yes                   |

Note: *IndepBoMemb* is board independence; *BoGenDivers* is board gender diversity; *CSRCom* is the presence of CSR committee; *BoSize* is board size; *CEOChairDual* is CEO chairman duality; *NBoMeet* is number of board meetings; *AudBoCom* is audit board committee; *BoSpecSkill* is board specific skills; *AveBoTen* is average board tenure; *NoExecBoMemb* is non-executive board members; *CSRSustExternAud* is external assurance; *StakeEngag* is stakeholder engagement; *size* is firm size; *Lev* is leverage ratio. *t*-statistics in parentheses.

\*\*\**p* < 0.01.

\*\**p* < 0.05.

\**p* < 0.1.

With respect to Hypothesis 3, the results reveal that there is a negative and statistically significant association between *CSRCom* and *EnvDecoup* across the four models. Therefore, the presence of a CSR committee reduces the environmental gap between environmental disclosure and real corporate performance. The negative coefficient suggests that the presence of a CSR committee is associated with a decrease in environmental decoupling. This finding reinforces the view that board dedicated committees provide valuable specialized perspective for strong oversight, which is represented in our study by environmental accountability and reducing the risk of decoupling. According to the legitimacy theory, board sustainability committees help firms in adhering to environmental regulations as well as stakeholder expectations. Their role in monitoring sustainability strategies can contribute to more credible and transparent corporate reporting. This finding is consistent with the result of Gull, Hussain, Khan, Khan, and Saeed (2023), who found that the establishment of a CSR committee, which is responsible for monitoring CSR issues, is an effective corporate governance mechanism that promotes lower levels of CSR misalignment.

With respect to the control variables, the findings in Model 4 reveal that *NBoMeet* is positively and statistically related to *EnvDecoup* ( $\beta = 0.001$ ,  $p < 0.1$ ), demonstrating that the higher the number of board meetings, the greater the environmental gap. This result is congruent with the literature arguing that board frequently meetings are an effective mechanism in the supervisory role of the board (Frias-Aceituno et al. 2013; Prado-Lorenzo and Garcia-Sanchez 2010; Vafeas 1999). In addition, the results reveal that *AveBoTen* is statistically and negatively related to *EnvDecoup* ( $\beta = -0.006$ ,  $p < 0.001$ ); thus, environmental misalignment decreases as the tenure of board members increases. This result confirms the argument made in previous studies (Patro et al. 2018), namely, that as their tenure increases, board members become more effective in managing environmental issues because they acquire more job experience and a better knowledge of the company's characteristics. The other variables used to measure board effectiveness are not statistically significant, similar to the findings of García-Sánchez et al. (2022).

In summary, the findings show strong empirical evidence on the role of board characteristics in mitigating environmental decoupling, specifically after the introduction of Directive 2014/95/EU. These findings highlight the importance of strong internal governance mechanisms in aligning environmental disclosure with performance and support EU policies that emphasize board structure and sustainability oversight.

### 4.3 | Additional Analyses

To shed light on the effects of board independence, board gender diversity, and the presence of a CSR committee on environmental decoupling, we performed additional analyses using the components of *EnvDecoup*, namely, *EnvDisc* and *EnvPerf*, as a dependent variable. Therefore, we re-estimated the panel regression with fixed effects by replacing the environmental decoupling variable with both environmental disclosure

(*EnvDisc*) and environmental performance (*EnvPerf*). We also re-estimated the four models and replaced the dependent variable with both environmental disclosure and environmental performance.

The results of these additional analyses are reported in Tables 6 and 7. More specifically, Table 6 shows the results using environmental disclosure as a dependent variable, whereas Table 7 reports the results using environmental performance as a dependent variable.

The results in Table 6 reveal that the higher the percentage of independent directors (*IndepBoMemb*) sitting on the board, the higher the environmental disclosure released by companies. However, the results are at the 10% significance level and are not consistent across all models. This result is similar to those of previous studies showing that a greater number of independent directors promote higher levels of CSR transparency (Gerged et al. 2021; Katmon et al. 2019; Pucheta-Martínez and López-Zamora 2018; Zaid et al. 2020).

With respect to board gender diversity, Table 6 Model 4 shows that the coefficient of *BoGenDivers* is positive; however, it is not statistically significant in all models for *EnvDisc*. Therefore, the findings reveal that the presence of women directors on the board is not always an effective mechanism for enhancing the environmental disclosure released by companies. This result is not aligned with those of previous studies (García-Sánchez et al. 2014; Hussain et al. 2018; Injeni et al. 2021; Liao et al. 2015), which may demonstrate that women directors enhance the level of CSR disclosure due to their characteristics.

The results of Table 6 Model 4 also reveal that the presence of a CSR committee positively and significantly influences environmental disclosure. The coefficient of *CSRCom* is positive and statistically significant for *EnvDisc* across the four models. Therefore, the establishment of a CSR committee responsible for overseeing CSR issues promotes the environmental information released by companies, in line with the theoretical arguments of agency theory (Gull, Hussain, Khan, Khan, and Saeed 2023). This result is also aligned with previous studies (Giannarakis et al. 2020; Helfaya and Moussa 2017; Peters and Romi 2014), which have found that the existence of a CSR committee increases the environmental information released.

Table 7 Model 4 reports the results for the effects of board independence, board gender diversity, and the presence of a CSR committee on environmental performance. The findings reveal that all three independent variables are positively and significantly related to environmental performance. More specifically, the coefficient of *IndepBoMemb* is positive and statistically significant for *EnvPerf* ( $\beta = 0.000$ ,  $p < 0.01$ ), revealing that the higher the number of independent directors sitting on the board, the greater the environmental performance achieved by companies, consistent with the findings of previous studies (e.g., Agarwala et al. 2023). At the same time, the coefficient of *BoGenDivers* is positive and statistically significant for *EnvPerf* across all models. This finding shows that the presence of women directors on the board promotes the achievement of

**TABLE 6** | Regression results (dependent variable *EnvDisc*).

| Variables               | Environmental disclosure |           |          |          |
|-------------------------|--------------------------|-----------|----------|----------|
|                         | Model 1                  | Model 2   | Model 3  | Model 4  |
| <i>IndepBoMemb</i>      | 0.000*                   | 0.000     | 0.000*   | 0.000    |
|                         | (1.678)                  | (0.522)   | (1.659)  | (1.386)  |
| <i>BoGenDivers</i>      | 0.001***                 | 0.001***  | 0.000    | 0.000    |
|                         | (8.887)                  | (7.200)   | (0.793)  | (0.610)  |
| <i>CSRCom</i>           | 0.048***                 | 0.033***  | 0.017*** | 0.017*** |
|                         | (10.517)                 | (7.252)   | (4.109)  | (2.916)  |
| <i>BoSize</i>           |                          | 0.001     | 0.001    | 0.001    |
|                         |                          | (0.748)   | (1.631)  | (1.319)  |
| <i>CEOChairDual</i>     |                          | −0.002    | 0.004    | 0.004    |
|                         |                          | (−0.342)  | (0.880)  | (0.640)  |
| <i>NBoMeet</i>          |                          | 0.001**   | 0.000    | 0.000    |
|                         |                          | (2.048)   | (0.623)  | (0.655)  |
| <i>AudBoCom</i>         |                          | 0.004     | −0.003   | −0.003   |
|                         |                          | (0.421)   | (−0.391) | (−0.574) |
| <i>BoSpecSkill</i>      |                          | 0.000     | 0.000    | 0.000    |
|                         |                          | (1.026)   | (0.531)  | (0.402)  |
| <i>AveBoTen</i>         |                          | 0.003***  | 0.001    | 0.001    |
|                         |                          | (2.903)   | (0.958)  | (0.582)  |
| <i>NoExecBoMemb</i>     |                          | 0.001***  | −0.000   | −0.000   |
|                         |                          | (4.192)   | (−1.275) | (−1.068) |
| <i>CSRSustExternAud</i> |                          | 0.008     | −0.001   | −0.001   |
|                         |                          | (1.271)   | (−0.230) | (−0.169) |
| <i>StakeEngag</i>       |                          | 0.035***  | 0.019*** | 0.019**  |
|                         |                          | (5.316)   | (3.241)  | (2.069)  |
| <i>size</i>             |                          | 0.055***  | 0.009*   | 0.009    |
|                         |                          | (10.745)  | (1.795)  | (1.329)  |
| <i>Leverage</i>         |                          | −0.039**  | −0.015   | −0.015   |
|                         |                          | (−2.098)  | (−0.884) | (−0.672) |
| Constant                | 0.164***                 | −1.182*** | 0.067    | 0.067    |
|                         | (18.215)                 | (−10.621) | (0.584)  | (0.428)  |
| Observations            | 3061                     | 3061      | 3061     | 3061     |
| R-squared               | 0.085                    | 0.161     | 0.359    | 0.359    |
| Number of firms         | 728                      | 728       | 728      | 728      |
| Control variables       | No                       | Yes       | Yes      | Yes      |
| Year FE                 | No                       | No        | Yes      | Yes      |
| Industry FE             | No                       | No        | No       | Yes      |
| Country FE              | No                       | No        | No       | Yes      |
| Cluster                 | No                       | No        | No       | Yes      |

\*\*\* $p < 0.01$ .\*\* $p < 0.05$ .\* $p < 0.1$ .

**TABLE 7** | Regression results (dependent variable *EnvPerf*).

| Variables               | Environmental performance |                       |                      |                     |
|-------------------------|---------------------------|-----------------------|----------------------|---------------------|
|                         | Model 1                   | Model 2               | Model 3              | Model 4             |
| <i>IndepBoMemb</i>      | 0.000***<br>(2.733)       | 0.000<br>(1.601)      | 0.000***<br>(2.828)  | 0.000**<br>(2.294)  |
| <i>BoGenDivers</i>      | 0.002***<br>(9.057)       | 0.001***<br>(7.464)   | 0.000**<br>(2.429)   | 0.000*<br>(1.936)   |
| <i>CSRCom</i>           | 0.068***<br>(13.236)      | 0.053***<br>(10.260)  | 0.038***<br>(7.766)  | 0.038***<br>(6.122) |
| <i>BoSize</i>           |                           | −0.001<br>(−1.047)    | −0.001<br>(−0.855)   | −0.001<br>(−0.714)  |
| <i>CEOChairDual</i>     |                           | −0.007<br>(−1.177)    | −0.003<br>(−0.477)   | −0.003<br>(−0.371)  |
| <i>NBoMeet</i>          |                           | 0.001**<br>(2.382)    | −0.000<br>(−0.042)   | −0.000<br>(−0.042)  |
| <i>AudBoCom</i>         |                           | 0.008<br>(0.739)      | 0.004<br>(0.384)     | 0.004<br>(0.351)    |
| <i>BoSpecSkill</i>      |                           | −0.000<br>(−1.038)    | −0.000<br>(−0.958)   | −0.000<br>(−0.873)  |
| <i>AveBoTen</i>         |                           | 0.004***<br>(3.573)   | 0.002**<br>(2.358)   | 0.002**<br>(2.038)  |
| <i>NoExecBoMemb</i>     |                           | 0.001**<br>(2.508)    | −0.000<br>(−1.451)   | −0.000<br>(−1.218)  |
| <i>CSRSustExternAud</i> |                           | 0.013*<br>(1.795)     | 0.005<br>(0.721)     | 0.005<br>(0.599)    |
| <i>StakeEngag</i>       |                           | 0.050***<br>(6.705)   | 0.036***<br>(5.247)  | 0.036***<br>(4.501) |
| <i>size</i>             |                           | 0.044***<br>(7.637)   | 0.002<br>(0.263)     | 0.002<br>(0.227)    |
| <i>Leverage</i>         |                           | −0.063***<br>(−3.027) | −0.044**<br>(−2.201) | −0.044*<br>(−1.892) |
| Constant                | 0.230***<br>(22.759)      | −0.848***<br>(−6.731) | 0.287**<br>(2.078)   | 0.287*<br>(1.769)   |
| Observations            | 3061                      | 3061                  | 3061                 | 3061                |
| R-squared               | 0.112                     | 0.171                 | 0.282                | 0.282               |
| Number of firms         | 728                       | 728                   | 728                  | 728                 |
| Control variables       | No                        | Yes                   | Yes                  | Yes                 |
| Year FE                 | No                        | No                    | Yes                  | Yes                 |
| Industry FE             | No                        | No                    | No                   | Yes                 |
| Country FE              | No                        | No                    | No                   | Yes                 |
| Cluster                 | No                        | No                    | No                   | Yes                 |

Note: *t*-statistics in parentheses.\*\*\**p* < 0.01.\*\**p* < 0.05.\**p* < 0.1.

higher environmental performance, congruent with the findings of previous studies (Atif et al. 2020; Post et al. 2011). Finally, the coefficient of *CSRCom* is positive and statistically significant for *EnvPerf*, showing that the presence of a CSR committee within the board is an effective governance mechanism that promotes the achievement of higher environmental performance, consistent with previous studies (Godos-Díez et al. 2018; Govindan et al. 2021; Orazalin 2019).

Taken together, the results of these additional analyses reveal that the presence of a CSR committee is an effective mechanism that promotes the release of a higher level of environmental information, the achievement of higher environmental performance, and, consequently, a reduction in environmental decoupling. Board independence and the presence of women on the board are effective mechanisms for increasing the environmental performance achieved by companies; however, they do not influence the quantity of environmental information released. Therefore, board gender diversity tends to only weakly reduce the environmental gap, thanks to its capacity for influencing environmental performance.

The results offer important insights into how these board characteristics affect corporate sustainability practices. The additional analysis focuses on symbolic actions reflected by environmental disclosure and substantive actions reflected by environmental performance. By examining these two components side by side, we gain a better understanding of the conditions under which environmental decoupling may occur, particularly when firms disclose information without matching improvements in actual performance.

In addition, corporate governance studies have often faced criticism regarding issues related to endogeneity (Al-Gamrh et al. 2024). Therefore, we reran our main results using 2SLS as well as generalized methods of moments (GMM) estimations to enhance the robustness to our main results. First, 2SLS addresses endogeneity by using instrumental variables, thereby creating a quasi-experimental setup that provides consistent estimation of results even if endogeneity is present. Table 8 shows our results using 2SLS, which are similar to our main findings. We tested the validity of our instruments using the Sargan and Basman tests for overidentifying restrictions. The results indicate no correlation between our instruments and error terms, thus validating our results. The results hold for all models with regard to the negative association between the presence of a CSR committee and environmental decoupling. However, the results tend to be slightly weaker (stronger) with regard to the negative impact of board independence and gender diversity on decoupling, particularly when including the control variables.

Second, we employed GMM models as supplementary estimation to validate the robustness of the main findings. The GMM approach enhances robustness by addressing endogeneity through the inclusion of lagged dependent variables. Table 9 shows the results of the GMM estimations incorporating various country, year, and industry dummies. The results are similar to the main results with slight variations across models, supporting the reliability of the main findings. The validity of the GMM regressions and instruments is evident by diagnostic

tests. The Arellano–Bond tests for serial correlation confirm the absence of second order serial correlation, whereas the Sargan test for overidentification demonstrates the validity of our instruments. The GMM results address any endogeneity concerns and confirm that the main findings are not driven by biases or omissions.

## 5 | Discussion, Implications, and Conclusions

This study aims to investigate the relationship between certain corporate governance characteristics and environmental decoupling. In the literature, several studies (e.g., García-Sánchez et al. 2022; Hawn and Ioannou 2016; Sauerwald and Su 2019; Tashman et al. 2019) have pointed out that companies often exhibit a misalignment between what they communicate externally and their internal practice. The existence of this gap can reduce the credibility and reliability of the environmental information released (Boiral et al. 2019), thus damaging the company's reputation (Du 2015), the value of the firm (Hawn and Ioannou 2016), and its capacity to access finance (García-Sánchez et al. 2021).

The findings of this study reveal that the presence of independent directors, as well as the inclusion of women board members and the presence of a CSR committee, fosters a reduction in environmental decoupling. Aligned with the agency theory framework, the results of this study show that the identified corporate governance characteristics are effective mechanisms in decreasing the misalignment between environmental disclosure and environmental performance that enhance the credibility of the environmental information released.

In addition, this paper shows that board independence and the presence of a CSR committee are mechanisms effective in enhancing both the environmental disclosures released by companies and the environmental performance achieved. Moreover, the presence of women directors has a positive effect on environmental performance; however, it does not influence the quantity of the environmental information provided. This finding, in contrast with the findings of previous studies (García-Sánchez et al. 2014; Hussain et al. 2018; Injen et al. 2021; Liao et al. 2015), demonstrates that women board members have no impact on the environmental information released. However, they do have the capacity to affect environmental performance. Therefore, women's characteristics—such as being more supportive, respectful, kind, and empathetic than men—lead them to be more oriented toward the needs of stakeholders and to be less inclined to accept opportunistic managerial behavior (Krishnan and Parsons 2008; Thorne et al. 2003), thereby increasing the environmental performance achieved.

This paper contributes to the literature in several ways. First, this research provides empirical evidence regarding the impact of different corporate governance characteristics on decoupling by investigating the environmental dimension, rather than the wider CSR concept. Therefore, this research contributes to the limited empirical analysis on the determinants of the CSR gap by demonstrating the effectiveness of

**TABLE 8** | Regression results using 2SLS (dependent variable *EnvDecoup*).

| Variables               | Environmental decoupling (2SLS) |                       |                       |                       |                       |
|-------------------------|---------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                         | Model 1                         | Model 2               | Model 3               | Model 4               | Model 5               |
| <i>IndepBoMemb</i>      | −0.004***<br>(−4.894)           |                       |                       | −0.002<br>(−1.194)    | 0.001<br>(1.158)      |
| <i>BoGenDivers</i>      |                                 | −0.011***<br>(−4.364) |                       | −0.015**<br>(−2.368)  | −0.009**<br>(−2.008)  |
| <i>CSRCom</i>           |                                 |                       | −0.148***<br>(−5.876) | −0.304***<br>(−2.709) | −0.376***<br>(−3.633) |
| <i>BoSize</i>           |                                 |                       |                       | −0.003<br>(−0.722)    | 0.004<br>(1.422)      |
| <i>CEOChairDual</i>     |                                 |                       |                       | 0.083***<br>(2.974)   | 0.031***<br>(3.031)   |
| <i>NBoMeet</i>          |                                 |                       |                       | 0.001<br>(0.725)      | 0.001<br>(0.769)      |
| <i>AudBoCom</i>         |                                 |                       |                       | 0.051<br>(1.414)      | −0.036<br>(−1.443)    |
| <i>BoSpecSkill</i>      |                                 |                       |                       | 0.000<br>(1.042)      | −0.000<br>(−0.674)    |
| <i>AveBoTen</i>         |                                 |                       |                       | −0.009***<br>(−3.370) | −0.008***<br>(−3.388) |
| <i>NoExecBoMemb</i>     |                                 |                       |                       | 0.001<br>(1.636)      | 0.000<br>(0.297)      |
| <i>CSRSustExternAud</i> |                                 |                       |                       | 0.099***<br>(2.699)   | 0.041***<br>(2.719)   |
| <i>StakeEngag</i>       |                                 |                       |                       | 0.037<br>(1.488)      | 0.042**<br>(1.993)    |
| <i>size</i>             |                                 |                       |                       | 0.017**<br>(2.180)    | 0.012**<br>(2.359)    |
| <i>Lev</i>              |                                 |                       |                       | −0.003<br>(−0.100)    | −0.029<br>(−0.704)    |
| Constant                | 0.169***<br>(3.552)             | 0.329***<br>(3.657)   | 0.062***<br>(2.899)   | 0.207<br>(1.627)      | 0.171<br>(1.098)      |
| Observations            | 3061                            | 3061                  | 3061                  | 3061                  | 3061                  |
| Control variables       | No                              | No                    | No                    | Yes                   | Yes                   |
| Year FE                 | No                              | No                    | No                    | No                    | Yes                   |
| Industry FE             | No                              | No                    | No                    | No                    | Yes                   |
| Country FE              | No                              | No                    | No                    | No                    | Yes                   |

\*\*\* $p < 0.01$ .\*\* $p < 0.05$ .

**TABLE 9** | Regression results using GMM (dependent variable *EnvDecoup*).

| Variables                    | Environmental decoupling (GMM) |                       |                       |                       |
|------------------------------|--------------------------------|-----------------------|-----------------------|-----------------------|
|                              | Model 1                        | Model 2               | Model 3               | Model 4               |
| <i>EnvDecoup = Log</i>       | −0.131**<br>(−1.977)           | −0.140**<br>(−2.154)  | −0.101*<br>(−1.676)   | −0.103*<br>(−1.660)   |
| <i>IndepBoMemb</i>           | −0.003***<br>(−3.214)          | −0.003***<br>(−3.634) | −0.003***<br>(−4.512) | −0.004***<br>(−4.147) |
| <i>BoGenDivers</i>           | −0.003***<br>(−4.437)          | −0.003***<br>(−3.795) | −0.001**<br>(−1.961)  | −0.001<br>(−1.560)    |
| <i>CSRCom</i>                | −0.080*<br>(−1.880)            | −0.091*<br>(−1.946)   | −0.082<br>(−1.626)    | −0.141***<br>(−3.134) |
| <i>BoSize</i>                | −0.003<br>(−0.545)             | −0.004<br>(−0.624)    | −0.009*<br>(−1.835)   | −0.010**<br>(−2.101)  |
| <i>CEOChairDual</i>          | 0.111***<br>(2.894)            | 0.092***<br>(2.584)   | 0.091***<br>(3.029)   | 0.103**<br>(2.435)    |
| <i>NBoMeet</i>               | 0.000<br>(0.187)               | −0.000<br>(−0.239)    | −0.002<br>(−1.005)    | −0.001<br>(−0.894)    |
| <i>AudBoCom</i>              | 1.641***<br>(4.639)            | 1.661***<br>(4.596)   | 1.625***<br>(5.376)   | 1.242***<br>(3.704)   |
| <i>BoSpecSkill</i>           | 0.002<br>(0.421)               | 0.004<br>(0.734)      | −0.001<br>(−0.291)    | 0.005<br>(1.094)      |
| <i>AveBoTen</i>              | −0.003<br>(−0.099)             | −0.009<br>(−0.341)    | 0.019<br>(0.869)      | 0.010<br>(0.403)      |
| <i>NoExecBoMemb</i>          | 0.015***<br>(3.003)            | 0.013***<br>(2.844)   | 0.011*<br>(1.944)     | 0.021***<br>(2.613)   |
| <i>CSRSustExternAud</i>      | 0.538**<br>(2.193)             | 0.464*<br>(1.899)     | 0.536**<br>(2.224)    | 0.581*<br>(1.787)     |
| <i>StakeEngag</i>            | −0.117<br>(−0.254)             | 0.127<br>(0.272)      | 0.175<br>(0.363)      | 0.360<br>(0.745)      |
| <i>size</i>                  | −0.056***<br>(−2.816)          | −0.047***<br>(−2.687) | −0.024<br>(−1.441)    | −0.015<br>(−0.682)    |
| <i>Leverage</i>              | 0.830**<br>(2.174)             | 1.194***<br>(2.594)   | 1.085**<br>(2.543)    | 1.098<br>(1.375)      |
| Constant                     | −2.022***<br>(−4.749)          | −2.385***<br>(−5.835) | −2.672***<br>(−6.141) | −3.659***<br>(−5.998) |
| Observations                 | 2259                           | 2259                  | 2259                  | 2259                  |
| Wald Chi2 ( <i>p</i> )       | 126.85(0.00)                   | 142.59(0.00)          | 200.37(0.00)          | 232.12(0.00)          |
| Number of firms              | 616                            | 616                   | 616                   | 616                   |
| Number of instruments        | 26                             | 43                    | 62                    | 67                    |
| Arellano–Bond AR(2) <i>p</i> | 0.548                          | 0.617                 | 0.999                 | 0.615                 |

(Continues)

TABLE 9 | (Continued)

| Variables            | Environmental decoupling (GMM) |         |         |         |
|----------------------|--------------------------------|---------|---------|---------|
|                      | Model 1                        | Model 2 | Model 3 | Model 4 |
| Sargan test <i>p</i> | 0.359                          | 0.526   | 0.469   | 0.527   |
| Industry FE          | No                             | Yes     | Yes     | Yes     |
| Country FE           | No                             | No      | Yes     | Yes     |
| Year FE              | No                             | No      | No      | Yes     |

\*\*\**p* < 0.01.\*\**p* < 0.05.\**p* < 0.1.

board independence, board gender diversity, and the presence of a CSR committee in monitoring opportunistic managerial behaviors and reducing the misalignment between disclosures and performance. This is also confirmed when investigating the environmental dimension exclusively. Second, this paper contributes to the agency theory and decoupling literature by providing empirical evidence regarding the association between corporate governance characteristics and decoupling in the mandatory sustainability disclosure setting. Our results show that although companies tend to underreport environmental data, the level of environmental decoupling decreased in 2023, thus demonstrating that the introduction of more stringent requirements for environmental disclosure (i.e., the Corporate Sustainability Reporting Directive 2022/2464/EU) could promote greater alignment between sustainability disclosures and performance.

This paper also has important practical implications. By revealing the presence of a misalignment between the environmental information released and the environmental performance, the paper recommends companies to pay attention to the coherence among the disclosure provided and the actions effectively implemented. To reduce the agency problems between agent and principal, in fact, it is not enough to increase the information released to external parties, but it is important to provide reliable disclosure. Therefore, companies should align their disclosure to the actions effectively implemented. In addition, by showing that the corporate governance characteristics identified reduce the environmental gap, this research recommends that companies design and configure the board of directors to align their environmental disclosure and performance. Moreover, the paper provides some public policy recommendations. First, by demonstrating that board independence and the presence of a CSR committee are mechanisms effective in enhancing the environmental disclosures released by companies and the environmental performance achieved, this paper provides suggestions for regulators and standard setters regarding how to structure boards to ensure that companies place a greater focus on CSR issues, in terms of both the disclosures released and the performance achieved. Second, by revealing that these board characteristics foster a reduction in environmental decoupling, the paper highlights the need for comprehensive rules and standards that boost true environmental practices and discourage the misalignment between the information released and the actions implemented, increasing the reliability of the disclosure

released. Finally, by revealing that the level of environmental decoupling decreased in 2023, after the introduction of stricter requirements for environmental disclosure, the paper provides useful suggestions to policymakers regarding the importance of regulating the sustainability disclosure released by companies. In 2023, the environmental disclosure increased for the sampled companies, probably in response to the Corporate Sustainability Reporting Directive 2022/2464/EU issued in December 2022 and the European Sustainability Reporting Standards issued by EFRAG. This result is particularly important considering the omnibus package issued in February 2025 that contains a proposal of simplification of the Corporate Sustainability Reporting Directive 2022/2464/EU. Among the main amending, the omnibus package suggests reducing the reporting burden by applying the reporting requirements exclusively to larger companies with more than 1000 employees. All the other companies may voluntarily disclose sustainability information referring to the Voluntary Standard for Micro and Small Enterprises (VSME) developed by EFRAG. In this context, the results of the paper showing an increase in environmental disclosure in 2023 provide positive support to the proposed process of amended Directive 2022/2464/EU.

Finally, this research has some limitations. First, although this study exclusively investigates the environmental dimension of sustainability, it would be interesting to analyze the social dimension. Thus, future studies could consider the social dimension and provide a better understanding of the presence/absence of misalignment between disclosure and performance and the role of corporate governance characteristics in reducing this gap. Moreover, this study exclusively investigates European listed companies that must respect European requirements regarding the sustainability information released. Future research could analyze companies operating in countries beyond the European context to understand the existence/absence of an environmental gap and the influence of corporate governance characteristics on this decoupling in settings where sustainability disclosures are voluntary or in those that have different sustainability disclosure requirements. Finally, this paper uses data from Refinitiv Eikon to perform the empirical analysis. However, it would be interesting to go further and submit questionnaires to or conduct interviews with board members to better understand how they really operate and make decisions regarding their companies' activities with respect to environmental and social issues.

## References

- Adams, R. B., and D. Ferreira. 2009. "Women in the Boardroom and Their Impact on Governance and Performance." *Journal of Financial Economics* 94, no. 2: 291–309. <https://doi.org/10.1016/j.jfineco.2008.10.007>.
- Agarwala, N., R. Pareek, and T. N. Sahu. 2023. "Does Board Independence Influence CSR Performance? A GMM-Based Dynamic Panel Data Approach." *Social Responsibility Journal* 19, no. 6: 1003–1022. <https://doi.org/10.1108/SRJ-10-2020-0433>.
- Al-Gamrh, B., U. Farooq, and T. Ahsan. 2024. "Beyond the Balance Sheet: Assessing Corporate Governance Through the Lens of Debtholders." *International Review of Financial Analysis* 96, no. Part A: 103625. <https://doi.org/10.1016/j.irfa.2024.103625>.
- Alfraih, M. M. 2016. "The Effectiveness of Board of Directors' Characteristics in Mandatory Disclosure Compliance." *Journal of Financial Regulation and Compliance* 24, no. 2: 154–176. <https://doi.org/10.1108/JFRC-07-2015-0035>.
- Arena C., R. Liong, and P. Vourvachis. 2018. "Carrot or Stick: CSR Disclosures by Southeast Asian Companies." *Sustainability Accounting, Management and Policy Journal* 9, no. 4: 422–454. <https://doi.org/10.1108/SAMPJ-06-2016-0037>.
- Armstrong, C. S., W. R. Guay, and J. P. Weber. 2010. "The Role of Information and Financial Reporting in Corporate Governance and Debt Contracting." *Journal of Accounting and Economics* 50, no. 2-3: 179–234. <https://doi.org/10.1016/j.jacceco.2010.10.001>.
- Atif, M., M. Hossain, M. S. Alam, and M. Goergen. 2020. "Does Board Gender Diversity Affect Renewable Energy Consumption?" *Journal of Corporate Finance* 66: 101665. <https://doi.org/10.1016/j.jcorpfin.2020.101665>.
- Barako, D. G., P. Hancock, and H. Y. Izan. 2006. "Factors Influencing Voluntary Corporate Disclosure by Kenyan Companies." *Corporate Governance: An International Review* 14, no. 2: 107–125. <https://doi.org/10.1111/j.1467-8683.2006.00491.x>.
- Beetsma, R., H. Peters, and E. Rebers. 2000. "When to Fire Bad Managers: The Role of Collusion Between Management and Board of Directors." *Journal of Economic Behavior and Organization* 42, no. 4: 427–444. [https://doi.org/10.1016/S0167-2681\(00\)00098-6](https://doi.org/10.1016/S0167-2681(00)00098-6).
- Boiral, O., I. Heras-Saizarbitoria, and M. C. Brotherton. 2019. "Assessing and Improving the Quality of Sustainability Reports: The Auditors' Perspective." *Journal of Business Ethics* 155: 703–721.
- Broman, G. I., and K. H. Robèrt. 2017. "A Framework for Strategic Sustainable Development." *Journal of Cleaner Production* 140: 17–31. <https://doi.org/10.1016/j.jclepro.2015.10.121>.
- Crilly, D., M. Zollo, and M. T. Hansen. 2012. "Faking It or Muddling Through? Understanding Decoupling in Response to Stakeholder Pressures." *Academy of Management Journal* 55, no. 6: 1429–1448. <https://doi.org/10.5465/amj.2010.0697>.
- De Colle, S., A. Henriques, and S. Sarasvathy. 2014. "The Paradox of Corporate Social Responsibility Standards." *Journal of Business Ethics* 125: 177–191.
- Du, X. 2015. "How the Market Values Greenwashing? Evidence From China." *Journal of Business Ethics* 128: 547–574.
- Du, K., and S. J. Wu. 2019. "Does External Assurance Enhance the Credibility of CSR Reports? Evidence From CSR-Related Misconduct Events in Taiwan." *Auditing: A Journal of Practice & Theory* 38, no. 4: 101–130. <https://doi.org/10.2308/ajpt-52418>.
- Engert, S., R. Rauter, and R. J. Baumgartner. 2016. "Exploring the Integration of Corporate Sustainability Into Strategic Management: A Literature Review." *Journal of Cleaner Production* 112: 2833–2850. <https://doi.org/10.1016/j.jclepro.2015.08.031>.
- European Commission. 2022. "Corporate Sustainability Reporting." [https://finance.ec.europa.eu/capital-marketsunion-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting\\_en](https://finance.ec.europa.eu/capital-marketsunion-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en).
- European Parliament and the Council. 2014. "Directive 2014/95/EU of the European Parliament and of the Council of 22 October 2014 Amending Directive 2013/34/EU as Regards Disclosure of Non-Financial and Diversity Information by Certain Large Undertakings and Groups." *Official Journal of the European Union* 330, no. 1: 1–9.
- Fama, E. F. 1980. "Banking in the Theory of Finance." *Journal of Monetary Economics* 6, no. 1: 39–57.
- Fama, E. F., and M. C. Jensen. 1983. "Agency Problems and Residual Claims." *Journal of Law and Economics* 26, no. 2: 327–349.
- Fernández-Gago, R., L. Cabeza-García, and M. Nieto. 2018. "Independent Directors' Background and CSR Disclosure." *Corporate Social Responsibility and Environmental Management* 25, no. 5: 991–1001. <https://doi.org/10.1002/csr.1515>.
- Frias-Aceituno, J. V., L. Rodriguez-Ariza, and I. M. Garcia-Sanchez. 2013. "The Role of the Board in the Dissemination of Integrated Corporate Social Reporting." *Corporate Social Responsibility and Environmental Management* 20, no. 4: 219–233. <https://doi.org/10.1002/csr.1294>.
- García-Meca, E., and J. P. Sanchez-Ballesta. 2010. "The Association of Board Independence and Ownership Concentration With Voluntary Disclosure: A Meta-Analysis." *European Accounting Review* 19, no. 3: 603–627. <https://doi.org/10.1080/09638180.2010.496979>.
- García-Sánchez, I. M., B. Cuadrado-Ballesteros, and C. Sepulveda. 2014. "Does Media Pressure Moderate CSR Disclosures by External Directors?" *Management Decision* 52, no. 6: 1014–1045. <https://doi.org/10.1108/MD-09-2013-0446>.
- García-Sánchez, I. M., N. Hussain, C. Aibar-Guzmán, and B. Aibar-Guzmán. 2022. "Assurance of Corporate Social Responsibility Reports: Does It Reduce Decoupling Practices?" *Business Ethics, the Environment and Responsibility* 31, no. 1: 118–138. <https://doi.org/10.1111/beer.12394>.
- García-Sánchez, I. M., N. Hussain, S. A. Khan, and J. Martínez-Ferrero. 2021. "Do Markets Punish or Reward Corporate Social Responsibility Decoupling?" *Business & Society* 60, no. 6: 1431–1467. <https://doi.org/10.1177/0007650319898839>.
- Geissdoerfer, M., D. Vladimirova, and S. Evans. 2018. "Sustainable Business Model Innovation: A Review." *Journal of Cleaner Production* 198: 401–416. <https://doi.org/10.1016/j.jclepro.2018.06.240>.
- Gennari, F., and D. M. Salvioni. 2019. "CSR Committees on Boards: The Impact of the External Country Level Factors." *Journal of Management and Governance* 23, no. 3: 759–785.
- Gerged, A. M., E. Beddewela, and C. J. Cowton. 2021. "Is Corporate Environmental Disclosure Associated With Firm Value? A Multicountry Study of Gulf Cooperation Council Firms." *Business Strategy and the Environment* 30, no. 1: 185–203. <https://doi.org/10.1002/bse.2616>.
- Giannarakis, G., A. Andronikidis, and N. Sariannidis. 2020. "Determinants of Environmental Disclosure: Investigating New and Conventional Corporate Governance Characteristics." *Annals of Operations Research* 294, no. 1: 87–105.
- Godos-Diez, J. L., L. Cabeza-García, D. Alonso-Martínez, and R. Fernández-Gago. 2018. "Factors Influencing Board of Directors' Decision-Making Process as Determinants of CSR Engagement." *Review of Managerial Science* 12: 229–253. <https://doi.org/10.1007/s11846-016-0220-1>.
- Govindan, K., M. Kilic, A. Uyar, and A. S. Karaman. 2021. "Drivers and Value-Relevance of CSR Performance in the Logistics Sector: A Cross-Country Firm-Level Investigation." *International Journal of Production Economics* 231: 107835. <https://doi.org/10.1016/j.ijpe.2020.107835>.

- Graafland, J., and H. Smid. 2019. "Decoupling Among CSR Policies, Programs, and Impacts: An Empirical Study." *Business & Society* 58, no. 2: 231–267. <https://doi.org/10.1177/0007650316647951>.
- Gul, F. A., B. Srinidhi, and J. S. Tsui. 2008. "Board Diversity and the Demand for Higher Audit Effort." Available at SSRN 1359450.
- Gull, A., N. Hussain, S. Khan, M. Nadeem, and A. Zalata. 2023. "Walking the Talk? A Corporate Governance Perspective on Corporate Social Responsibility Decoupling." *British Journal of Management* 34, no. 4: 2186–2211. <https://doi.org/10.1111/1467-8551.12695>.
- Gull, A. A., N. Hussain, S. A. Khan, Z. Khan, and A. Saeed. 2023. "Governing Corporate Social Responsibility Decoupling: The Effect of the Governance Committee on Corporate Social Responsibility Decoupling." *Journal of Business Ethics* 185, no. 2: 349–374. <https://doi.org/10.1007/s10551-022-05181-3>.
- Gull, A. A., N. Hussain, S. A. Khan, R. Mushtaq, and R. Orij. 2023. "The Power of the CEO and Environmental Decoupling." *Business Strategy and the Environment* 32, no. 6: 3951–3964. <https://doi.org/10.1002/bse.3347>.
- Hair, J., R. Anderson, R. Tatham, and W. Black. 1998. *Multivariate Data Analysis*. 5th ed. Prentice Hall.
- Hawn, O., and I. Ioannou. 2016. "Mind the Gap: The Interplay Between External and Internal Actions in the Case of Corporate Social Responsibility." *Strategic Management Journal* 37, no. 13: 2569–2588. <https://doi.org/10.1002/smj.2464>.
- Helfaya, A., and T. Moussa. 2017. "Do Boards' Corporate Social Responsibility Strategy and Orientation Influence Environmental Sustainability Disclosure? UK Evidence." *Business Strategy and the Environment* 26, no. 8: 1061–1077. <https://doi.org/10.1002/bse.1960>.
- Hillman, A., and T. Dalziel. 2003. "Boards of Directors and Firm Performance: Integrating Agency and Resource Dependence Perspectives." *Academy of Management Review* 28, no. 3: 383–396. <https://doi.org/10.5465/amr.2003.10196729>.
- Hussain, N., I. M. García-Sánchez, S. A. Khan, Z. Khan, and J. Martínez-Ferrero. 2023. "Connecting the Dots: Do Financial Analysts Help Corporate Boards Improve Corporate Social Responsibility?" *British Journal of Management* 34, no. 1: 363–389. <https://doi.org/10.1111/1467-8551.12586>.
- Hussain, N., U. Rigoni, and R. P. Orij. 2018. "Corporate Governance and Sustainability Performance: Analysis of Triple Bottom Line Performance." *Journal of Business Ethics* 149: 411–432. <https://doi.org/10.1007/s10551-016-3099-5>.
- Injeni, G., M. Mangena, D. Mathuva, and R. Mudida. 2021. "Agency and Institutional-Related Factors and the Heterogeneity of Sustainability and Integrated Report Information Disclosures in Kenya." *Journal of Financial Reporting and Accounting* 20, no. 5: 809–840. <https://doi.org/10.1108/jfra-10-2020-0305>.
- Jensen, M. C., and W. H. Meckling. 1976. "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure." *Journal of Financial Economics* 3, no. 4: 305–360.
- Katmon, N., Z. Z. Mohamad, N. M. Norwani, and O. A. Farooque. 2019. "Comprehensive Board Diversity and Quality of Corporate Social Responsibility Disclosure: Evidence From an Emerging Market." *Journal of Business Ethics* 157: 447–481. <https://doi.org/10.1007/s10551-017-3672-6>.
- Khan, Z., Y. K. Lew, and B. I. Park. 2015. "Institutional Legitimacy and Norms-Based CSR Marketing Practices: Insights From MNCs Operating in a Developing Economy." *International Marketing Review* 32, no. 5: 463–491.
- Klettner, A., T. Clarke, and M. Boersma. 2014. "The Governance of Corporate Sustainability: Empirical Insights Into the Development, Leadership and Implementation of Responsible Business Strategy." *Journal of Business Ethics* 122, no. 1: 145–165.
- Krishnan, G. V., and L. M. Parsons. 2008. "Getting to the Bottom Line: An Exploration of Gender and Earnings Quality." *Journal of Business Ethics* 78: 65–76. <https://doi.org/10.1007/s10551-006-9314-z>.
- Lepore, L., L. Landriani, S. Pisano, G. D'Amore, and S. Pozzoli. 2023. "Corporate Governance in the Digital Age: The Role of Social Media and Board Independence in CSR Disclosure. Evidence From Italian Listed Companies." *Journal of Management and Governance* 27: 749–785.
- Liao, L., L. Luo, and Q. Tang. 2015. "Gender Diversity, Board Independence, Environmental Committee and Greenhouse Gas Disclosure." *British Accounting Review* 47, no. 4: 409–424. <https://doi.org/10.1016/j.bar.2014.01.002>.
- Lock, I., and P. Seele. 2016. "The Credibility of CSR (Corporate Social Responsibility) Reports in Europe. Evidence From a Quantitative Content Analysis in 11 Countries." *Journal of Cleaner Production* 122: 186–200. <https://doi.org/10.1016/j.jclepro.2016.02.060>.
- Manninen, K., and J. Huiskonen. 2022. "Factors Influencing the Implementation of an Integrated Corporate Sustainability and Business Strategy." *Journal of Cleaner Production* 343: 131036. <https://doi.org/10.1016/j.jclepro.2022.131036>.
- Marquis, C., and C. Qian. 2014. "Corporate Social Responsibility Reporting in China: Symbol or Substance?" *Organization Science* 25, no. 1: 127–148. <https://doi.org/10.1287/orsc.2013.0837>.
- Matsa, D. A., and A. R. Miller. 2013. "A Female Style in Corporate Leadership? Evidence From Quotas." *American Economic Journal: Applied Economics* 5, no. 3: 136–169.
- Menard, S. 2002. *Applied Logistic Regression Analysis* (No. 106). Sage.
- Meuer, J., J. Koelbel, and V. H. Hoffmann. 2020. "On the Nature of Corporate Sustainability." *Organization & Environment* 33, no. 3: 319–341. <https://doi.org/10.1177/1086026619850180>.
- Muttakin, M. B., and N. Subramaniam. 2015. "Firm Ownership and Board Characteristics: Do They Matter for Corporate Social Responsibility Disclosure of Indian Companies?" *Sustainability Accounting, Management and Policy Journal* 6, no. 2: 138–165. <https://doi.org/10.1108/SAMPJ-10-2013-0042>.
- Orazalin, N. 2019. "Do Board Sustainability Committees Contribute to Corporate Environmental and Social Performance? The Mediating Role of Corporate Social Responsibility Strategy." *Business Strategy and the Environment* 29, no. 1: 140–153. <https://doi.org/10.1002/bse.2354>.
- Patro, S., L. Y. Zhang, and R. Zhao. 2018. "Director Tenure and Corporate Social Responsibility: The Tradeoff Between Experience and Independence." *Journal of Business Research* 93: 51–66.
- Peters, G. F., and A. M. Romi. 2014. "Does the Voluntary Adoption of Corporate Governance Mechanisms Improve Environmental Risk Disclosures? Evidence From Greenhouse Gas Emission Accounting." *Journal of Business Ethics* 125, no. 4: 637–666.
- Porter, M. E., and M. R. Kramer. 2011. "Creating Shared Value." *Harvard Business Review* 89: 62–77.
- Post, C., N. Rahman, and E. Rubow. 2011. "Green Governance: Boards of Directors' Composition and Environmental Corporate Social Responsibility." *Business & Society* 50, no. 1: 189–223. <https://doi.org/10.1177/0007650310394642>.
- Prado-Lorenzo, J. M., and I. M. Garcia-Sanchez. 2010. "The Role of the Board of Directors in Disseminating Relevant Information on Greenhouse Gases." *Journal of Business Ethics* 97: 391–424.
- Pucheta-Martínez, M. C., and B. López-Zamora. 2018. "Engagement of Directors Representing Institutional Investors on Environmental Disclosure." *Corporate Social Responsibility and Environmental Management* 25, no. 6: 1108–1120. <https://doi.org/10.1002/csr.1525>.
- Roberts, L., N. Georgiou, and A. M. Hassan. 2022. "Investigating Biodiversity and Circular Economy Disclosure Practices: Insights From

Global Firms.” *Corporate Social Responsibility and Environmental Management* 30, no. 3: 1053–1069. <https://doi.org/10.1002/csr.2402>.

Saeed, A., S. Hamid, R. Manita, and P. Suntraruk. 2025. “CSR Decoupling and Financial Fraud: Unveiling the Hidden Nexus in US-Listed Firms.” *Research in International Business and Finance* 75: 102791. <https://doi.org/10.1016/j.ribaf.2025.102791>.

Sauerwald, S., and W. Su. 2019. “CEO Overconfidence and CSR Decoupling.” *Corporate Governance: An International Review* 27, no. 4: 283–300.

Shahab, Y., A. A. Gull, T. Ahsan, and R. Mushtaq. 2022. “CEO Power and Corporate Social Responsibility Decoupling.” *Applied Economics Letters* 29, no. 21: 1965–1969. <https://doi.org/10.1080/13504851.2021.1966368>.

Sinclair-Desgagné, B., and E. Gozlan. 2003. “A Theory of Environmental Risk Disclosure.” *Journal of Environmental Economics and Management* 45: 377–393. [https://doi.org/10.1016/S0095-0696\(02\)00056-6](https://doi.org/10.1016/S0095-0696(02)00056-6).

Stewart, R., and M. Niero. 2018. “Circular Economy in Corporate Sustainability Strategies: A Review of Corporate Sustainability Reports in the Fast-Moving Consumer Goods Sector.” *Business Strategy and the Environment* 27, no. 7: 1005–1022. <https://doi.org/10.1002/bse.2048>.

Tashman, P., V. Marano, and T. Kostova. 2019. “Walking the Walk or Talking the Talk? Corporate Social Responsibility Decoupling in Emerging Market Multinationals.” *Journal of International Business Studies* 50: 153–171. <https://doi.org/10.1057/s41267-018-0171-7>.

Tata, J., and S. Prasad. 2015. “CSR Communication: An Impression Management Perspective.” *Journal of Business Ethics* 132: 765–778.

Thorne, L., D. W. Massey, and M. Magnan. 2003. “Institutional Context and Auditors’ Moral Reasoning: A Canada-US Comparison.” *Journal of Business Ethics* 43: 305–321. <https://doi.org/10.1023/A:1023005311277>.

Vafeas, N. 1999. “Board Meeting Frequency and Firm Performance.” *Journal of Financial Economics* 53, no. 1: 113–142.

Zaid, M. A., S. T. Abuhijleh, and M. C. Pucheta-Martínez. 2020. “Ownership Structure, Stakeholder Engagement, and Corporate Social Responsibility Policies: The Moderating Effect of Board Independence.” *Corporate Social Responsibility and Environmental Management* 27, no. 3: 1344–1360. <https://doi.org/10.1002/csr.1888>.

## Appendix I

### Environmental Disclosure and Environmental Performance Components of Environmental Decoupling

---

#### Environmental disclosure components

##### *Resource use*

1. Does the company report on initiatives to reduce, reuse, substitute, or phase out toxic chemicals or substances? (Toxic chemicals reduction)
2. Does the company claim to use environmental criteria (e.g., life cycle assessment) to source or eliminate materials? (Environmental materials sourcing)
3. Does the company report about environmentally friendly or green sites or offices? (Green buildings)
4. Does the company report on initiatives to reduce the environmental impact on land owned, leased, or managed for production activities or extractive use? (Land environmental impact reduction)
5. Does the company report or show to be ready to end a partnership with a sourcing partner, if environmental criteria are not met? (Env supply chain partnership termination)

##### *Innovation*

6. Does the company disclose the current share or amount of green capital expenditure deployed to climate-related opportunities (green capex)? (Green capex)
7. Has the company disclosed a target for the share or amount of expenditure deployed to climate-related opportunities (green capex)? (Green capex target)
8. Has the company disclosed a target to increase the share of green revenues in its overall sales? (Green revenues target)
9. Does the company report on at least one product line or service that is designed to have positive effects on the environment or which is environmentally labeled and marketed? (Environmental products)
10. Is the company a signatory of the Equator principles (commitment to manage environmental issues in project financing) or does it claim to evaluate projects on the basis of environmental or biodiversity risks as well? (Equator principles or Env project financing)
11. Does the company report on specific products which are designed for reuse, recycling or the reduction of environmental impacts? (Eco-design products)
12. Does the company claim to produce, source or distribute wood or forest products that are labeled (e.g., Forest Stewardship Council (FSC))? (Labeled Wood)
13. Does the company report or show initiatives to produce or promote organic food or other products? (Organic products initiatives)
14. Does the company report about take-back procedures and recycling programs to reduce the potential risks of products entering the environment or does the company report about product features or services that will promote responsible and environmentally preferable use? (Product impact minimization)
15. Does the company report about take-back procedures and recycling programs to reduce the potential risks of products entering the environment? (Take-back and recycling initiatives)
16. Does the company report about product features and applications or services that will promote responsible, efficient, cost-effective, and environmentally preferable use? (Product environmental responsible use)
17. Does the company claim to lease, rent or market buildings that are certified by BREEAM, LEED or any other nationally recognized real estate certification? (Real estate sustainability certifications)
18. Does the company report on assets under management which employ environmental screening criteria or environmental factors in the investment selection process? (Environmental assets under mgt)

##### *Emissions*

19. Does the company describe its process for identifying and assessing climate-related risks? (Climate-related risks assessment process)
  20. Does the company clarify the role and type of offsets/negative emission technologies in its transition plan? (Transition plan offsets)
  21. Does the company disclose assets or business activities vulnerable to transition risk? (Financial exposure to transition risk)
  22. Does the company disclose assets or business activities vulnerable to physical climate risk? (Financial exposure to physical risk)
  23. Does the company report the temperature alignment of its portfolios or lending activities? (Portfolio alignment)
  24. Does the company report on its impact on biodiversity or on activities to reduce its impact on the native ecosystems and species, as well as the biodiversity of protected and sensitive areas? (Biodiversity impact reduction)
  25. Does the company report on initiatives to reduce, reuse, recycle, substitute, or phase out SO<sub>x</sub> (sulfur oxides) or NO<sub>x</sub> (nitrogen oxides) emissions? (Nox and Sox emissions reduction)
  26. Does the company report on initiatives to reduce, substitute, or phase out volatile organic compounds (VOC) or particulate matter less than ten microns in diameter (PM<sub>10</sub>)? (Voc or particulate matter emissions reduction)
  27. Does the company report on initiatives to reduce, substitute, or phase out volatile organic compounds (VOC)? (Voc emission reduction)
  28. Does the company report on initiatives to reduce, substitute, or phase out particulate matter less than ten microns in diameter (PM<sub>10</sub>)? (Particulate matter emissions reduction)
  29. Does the company report on initiatives to recycle, reduce, reuse, substitute, treat, or phase out e-waste? (E-waste reduction)
  30. Does the company report on initiatives to recycle, reduce, reuse, substitute, treat, or phase out total waste? (Waste reduction initiatives)
  31. Does the company claim to have an ISO 14000 or EMS certification? (ISO 14000 or EMS)
  32. Does the company report or provide information on company-generated initiatives to restore the environment? (Environmental restoration initiatives)
  33. Does the company report on initiatives to reduce the environmental impact of transportation used for its staff? (Staff transportation impact reduction)
  34. Does the company report on its environmental expenditures or does the company report to make proactive environmental investments to reduce future risks or increase future opportunities? (Environmental expenditures investment)
  35. Does the company report on making proactive environmental investments or expenditures to reduce future risks or increase future opportunities? (Environmental investments initiatives)
  36. Does the company report on partnerships or initiatives with specialized NGOs, industry organizations, governmental or supra-governmental organizations, which are focused on improving environmental issues? (Environmental partnerships)
-

---

## Environmental performance components

### Resource use

1. Does the company set specific objectives to be achieved on resource efficiency? (Resource reduction targets)
2. Does the company have a policy for reducing the use of natural resources or to lessen the environmental impact of its supply chain? (Resource reduction policy)
3. Does the company have a policy to improve its water efficiency? (Policy water efficiency)
4. Does the company have a policy to improve its energy efficiency? (Policy energy efficiency)
5. Does the company have a policy to improve its use of sustainable packaging? (Policy sustainable packaging)
6. Does the company have a policy to include its supply chain in the company's efforts to lessen its overall environmental impact? (Policy environmental supply chain)
7. Does the company make use of renewable energy? (Renewable energy use)
8. Does the company use environmental criteria (ISO 14000, energy consumption, etc.) in the selection process of its suppliers or sourcing partners? (Environmental supply chain management)
9. Has the company set targets or objectives to be achieved on water efficiency? (Targets water efficiency)
10. Has the company set targets or objectives to be achieved on energy efficiency? (Targets energy efficiency)
11. Does the company train its employees on environmental issues? (Environmental management training)
12. Does the company have an environmental management team? (Environmental management team)

### Innovation

13. Does the company develop new products that are marketed as reducing noise emissions? (Noise reduction)
14. Is the company developing hybrid vehicles? (Hybrid vehicles)
15. Is the company a signatory of the Equator principles (commitment to manage environmental issues in project financing)? (Equator principles)
16. Does the company construct nuclear reactors, produce nuclear energy or is active in another way in the nuclear energy industry? (Nuclear)
17. Does the company produce or distribute genetically modified organisms (GMO) or seeds? (GMO products)
18. Does the company develop products or technologies for use in the clean, renewable energy (such as wind, solar, hydro, and geothermal and biomass power)? (Renewable clean energy products)
19. Does the company develop products or technologies that are used for water treatment, purification or that improve water use efficiency? (Water technologies)
20. Does the company develop products and services that improve the energy efficiency of buildings? (Sustainable building products)
21. Does the company produce or distribute agrochemicals like pesticides, fungicides or herbicides? (Agrochemical products)
22. Are the revenues generated by the company from agrochemicals like pesticides, fungicides, or herbicides 5% or more of company sales? (Agrochemical 5% revenue)
23. Is the company directly or indirectly involved in animal testing for cosmetics? (Animal testing cosmetic)
24. Has the company established a program or an initiative to reduce, phase out, or substitute for animal testing? (Animal testing reduction)

### Emissions

25. Does the transition plan detail how the company's transition efforts are reflected in its financial planning? (Transition plan financial planning)
  26. Does the transition plan address material Scope 3 emissions? (Transition plan Scope 3 emissions)
  27. Is the company aligned with the Paris agreement based on 7% reduction target of greenhouse gas emissions, Scope 1, Scope 2, and Scope 3? (GHG emissions scope 1, 2, and 3 Paris agreement)
  28. Does the company have a policy to improve emission reduction? (Policy emissions)
  29. Has the company set targets or objectives to be achieved on emission reduction? (Targets emissions)
  30. Does the company have an internal price on carbon? (Internal carbon pricing)
  31. Does the company have policy in place for nuclear safety and emergency preparedness to minimize the impact of nuclear disasters in and around nuclear plants? (Policy nuclear safety)
  32. Does the company conduct surveys of the environmental performance of its suppliers? (Environmental supply chain monitoring)
  33. Is the company aware that climate change can represent commercial risks and/or opportunities? (Climate change commercial risk opportunities)
  34. Is the company aligned with the Paris agreement based on 7% reduction target of greenhouse gas emissions, Scope 1 and Scope2? (GHG emissions scope 1,2 Paris agreement aligned)
  35. Is the company aligned with the Paris agreement based on 7% reduction target of greenhouse gas emissions, Scope 1, Scope 2, and Scope 3 to million revenues USD? (GHG emissions intensity scope 1, 2, and 3 Paris agreement aligned)
  36. Is the company aligned with the Paris agreement based on 7% reduction target of greenhouse gas emissions, scope 1 and 2 to million revenues USD? (GHG emissions intensity scope 1 and 2 Paris agreement aligned)
  37. Does the transition plan cover the short-term time horizon (next 5 years)? Medium-term time horizon (next 5–15 years)? Long-term time horizon (after the next 15 years)? (Transition plan time horizon coverage)
-